Volatility Forecasting Report

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Executive Summary

This report presents a comprehensive analysis of volatility forecasting using Heterogeneous Autoregressive (HAR) and HAR with exogenous variables (HAR-X) models applied to Treasury Bond ETF (TLT) data.

Key Objectives: - Evaluate multiple volatility estimators (Squared Return, Parkinson, Garman-Klass, Rogers-Satchell) - Compare HAR and HAR-X model performance across different rolling window sizes - Implement ensemble forecasting using inverse QLIKE weighting - Validate models using out-of-sample testing and statistical comparisons

Main Findings: - Ensemble models outperform individual estimators across all metrics - HAR-X with window=756 provides most stable and consistent predictions - Exogenous variables offer marginal but meaningful improvement in forecast calibration - No statistically significant difference between windows 504 and 756 (DM test) - Both models demonstrate strong forecasting ability with well-behaved residuals

Data Description

Dataset: iShares 20+ Year Treasury Bond ETF (TLT)
Period: 2003-01-01 to 2024-12-30
Frequency: Daily
Total Observations: 5536

Price Data Components: - Open, High, Low, Close prices - Trading volume - Adjusted close prices

Target Variable: - Realized Volatility (RV): Annualized variance computed from log returns - Log-transformed for modeling to ensure stationarity

Train/Test Split: - Training Set: 2003-01-06 to 2022-12-30 (data before 2023-01-01) — 5033 observations - Test Set: 2023-01-03 to 2024-12-30 (data on/after 2023-01-01) — 501 observations

Methodology

Model Framework

1. Volatility Estimation

Four volatility estimators are computed from OHLC data: - **Squared Return (RV)**: $^2 = 252 \times (\log(C\ /C\ ))^2$ - **Parkinson**: $^2 = 252 \times (1/(4\ln 2)) \times (\log(H\ /L\ ))^2$ - **Garman-Klass**: $^2 = 252 \times [0.5(\log(H\ /L\ ))^2 - (2\ln 2-1)(\log(C\ /O\ ))^2]$ - **Rogers-Satchell**: $^2 = 252 \times [\log(H\ /O\ )\log(H\ /C\ ) + \log(L\ /O\ )\log(L\ /C\ )]$

All estimators are log-transformed for modeling.

2. HAR Model

The HAR model captures heterogeneous volatility components:

$\log(RV\ ) = \ + \ \cdot RV\ + \ \cdot RV\ :\ + \ \cdot RV\ :\ +$

Where: - RV : Daily component (lag 1) - RV : : Weekly component (5-day average) - RV : : Monthly component (22-day average)

3. HAR-X Model

Extends HAR by adding exogenous variables:

log(RV) = + · RV + · RV : + · RV : + Σ · X +

Exogenous variables (X): - UST10Y (10-Year Treasury Yield) - HYOAS (High Yield Spread) - TermSpread (10Y-2Y) - VIX (Volatility Index) - Breakeven10Y (Inflation expectations)

4. Rolling Window Estimation

Models estimated using rolling windows: 252, 504, 756, 1008, 1260 days

5. Ensemble Forecasting

Predictions combined using inverse QLIKE weighting:

w = (1/QLIKE) / Σ (1/QLIKE)

Final forecast: ŷ = Σ w × ŷ

6. Evaluation Metrics

- **QLIKE**: log(ˆ²) + ² /ˆ² (forecast calibration)
- **MSPE**: ((² - ˆ²)/ ²)² (percentage error)
- **RMSE**: √(E[(² - ˆ²)²]) (absolute error)
- **Diebold-Mariano Test**: Statistical comparison of forecast accuracy
- **Ljung-Box Test**: Residual autocorrelation check

Volatility Estimators Analysis

Pre-Model Diagnostics

The following table presents the stationarity and autocorrelation diagnostics for each volatility estimator. We use the Augmented Dickey-Fuller (ADF) test and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test to assess stationarity, and the Ljung-Box test to check for serial correlation in the residuals.

Table 1: Diagnostic Tests for Volatility Estimators (Part 1/3)

Estimator	ADF stat	ADF p	ADF pass (p)	KPSS stat
square_est_log	-9.175	0	True	1.852
parkinson_est_log	-5.068	0	True	0.876
gk_est_log	-4.993	0	True	0.927
rs_est_log	-5.862	0	True	1.251

Table 1: Diagnostic Tests for Volatility Estimators (Part 2/3)

Estimator	KPSS p	KPSS pass (p>)	LB p @10	LB p @20
square_est_log	0.01	False	0	0

Estimator	KPSS p	KPSS pass (p>)	LB p @10	LB p @20
parkinson_est_log	0.01	False	0	0
gk_est_log	0.01	False	0	0
rs_est_log	0.01	False	0	0

Table 1: Diagnostic Tests for Volatility Estimators (Part 3/3)

Estimator	White noise (LB)	Stationary (ADF KPSS)
square_est_log	0	0
parkinson_est_log	0	0
gk_est_log	0	0
rs_est_log	0	0

ACF and PACF Analysis

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots reveal: - **Slow decay in ACF**: Indicates long memory in volatility, consistent with volatility clustering - **Significant PACF spikes**: Suggests short-term AR effects up to 5-15 lags - **HAR model justification**: These patterns support using daily (1), weekly (5), and monthly (22) lags

ACF and PACF for All Volatility Estimators

HAR Model Results

Performance Metrics Across Windows

The HAR model was evaluated across multiple rolling window sizes: 252, 504, 756, 1008, and 1260 days. Below are the comprehensive performance metrics for each estimator and window size.

Table 2: HAR Model Performance Summary (QLIKE and MSPE)

	QLIKE_mean	QLIKE_std	MSPE_mean	MSPE_std
(252, 'square_est_log')	-0.769	12.111	1.41257e+17	3.76377e+18
(252, 'parkinson_est_log')	-1.198	8.147	1.6656e+17	4.56201e+18
(252, 'gk_est_log')	-1.162	8.635	1.90755e+17	6.10036e+18
(252, 'rs_est_log')	-1.092	9.508	2.15613e+17	7.36021e+18
(504, 'square_est_log')	-0.91	9.818	1.30743e+17	3.60831e+18
(504, 'parkinson_est_log')	-1.167	7.949	1.2193e+17	2.64853e+18
(504, 'gk_est_log')	-1.161	7.916	1.21592e+17	2.70329e+18
(504, 'rs_est_log')	2.677	241.793	9.89925e+16	1.90839e+18

	QLIKE_mean	QLIKE_std	MSPE_mean	MSPE_std
(756, 'square_est_log')	-0.712	11.015	1.20577e+17	2.87891e+18
(756, 'parkinson_est_log')	-1.074	8.005	1.13032e+17	2.38754e+18
(756, 'gk_est_log')	-1.079	7.85	1.11258e+17	2.41615e+18
(756, 'rs_est_log')	-0.548	18.87	8.77846e+16	1.62212e+18
(1008, 'square_est_log')	-0.659	10.463	1.34258e+17	2.59699e+18
(1008, 'parkinson_est_log')	-1.024	7.822	1.37834e+17	2.82278e+18
(1008, 'gk_est_log')	-1.022	7.723	1.33777e+17	2.76729e+18
(1008, 'rs_est_log')	-0.587	16.067	1.07053e+17	1.98173e+18
(1260, 'square_est_log')	-0.702	10.034	1.42917e+17	2.54673e+18
(1260, 'parkinson_est_log')	-0.999	8.011	1.58539e+17	3.11109e+18
(1260, 'gk_est_log')	-1.02	7.764	1.54594e+17	3.04031e+18
(1260, 'rs_est_log')	-0.666	15.929	1.22436e+17	2.10492e+18

Table 3: Ljung-Box Test Results for HAR Model Residuals

	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(252, 'lb_stat_10')	16.0084759385020	8.871809235100	8.767704642792	15.592590450654132
(252, 'lb_p_10')	0.0993900239809	0.4188929290750	0.7553385460346	0.9532475306937539
(252, 'lb_stat_20')	21.8194150034171	13.1628444119371	15.2006099219339	18.7821908008865267
(252, 'lb_p_20')	0.3503895516427	0.3868824076348	0.7760585078169	0.5399627776490109
(252, 'white_noise_flag')	True	True	True	True
(252, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(252, 'n_obs')	4759	4759	4759	4759
(252, 'name')	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(504, 'lb_stat_10')	15.0184911316007	7.838550988728178	7.825067072940083	13.52359088932283
(504, 'lb_p_10')	0.1313891558874	0.5687617725958	0.7971587096276	0.3958171664521069
(504, 'lb_stat_20')	20.9101629912546	16.4246590680508	13.2820356364909	10.907925612625665
(504, 'lb_p_20')	0.4024372413949	0.8966554935787	0.5797655449846	0.1658894409312155
(504, 'white_noise_flag')	True	True	True	True

	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(504, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(504, 'n_obs')	4507	4507	4507	4507
(504, 'name')	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(756, 'lb_stat_10')	20.540204095881	8.0778522065617	7.24685843441579	15.533288133808632
(756, 'lb_p_10')	0.0245379690497	0.7569614640108	0.58316554740509	0.9953634026038526
(756, 'lb_stat_20')	24.523774745157	13.2219989114353	10.7297642374951	17.879330668939435
(756, 'lb_p_20')	0.2202608247432	0.4938295978023	0.9827940623884	0.5285357139142221
(756, 'white_noise_flag')	False	True	True	True
(756, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(756, 'n_obs')	4255	4255	4255	4255
(756, 'name')	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(1008, 'lb_stat_10')	23.2950113332672	12.0566863022551	16.7086873346665	15.97219370618098
(1008, 'lb_p_10')	0.0097084905201	0.4385977901364	0.924584157839	0.3514889532164634
(1008, 'lb_stat_20')	27.727526338080	13.7524245119941	14.5305583524819	19.8439201894487
(1008, 'lb_p_20')	0.1160079273748	0.8583844883298	0.8039651574903	0.4677305859141015
(1008, 'white_noise_flag')	False	True	True	True
(1008, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(1008, 'n_obs')	4003	4003	4003	4003
(1008, 'name')	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(1260, 'lb_stat_10')	19.244397173843	12.6743875226794	17.8516504980317	16.672777441706476
(1260, 'lb_p_10')	0.0372660519826	0.9378176009360	0.2961077063784	0.08179254893235461
(1260, 'lb_stat_20')	24.352538087781	19.8724615035366	16.5187756879457	21.90072582917874
(1260, 'lb_p_20')	0.2273316919990	0.4919423438067	0.7095548849646	0.3459217478471632
(1260, 'white_noise_flag')	False	True	True	True
(1260, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(1260, 'n_obs')	3751	3751	3751	3751

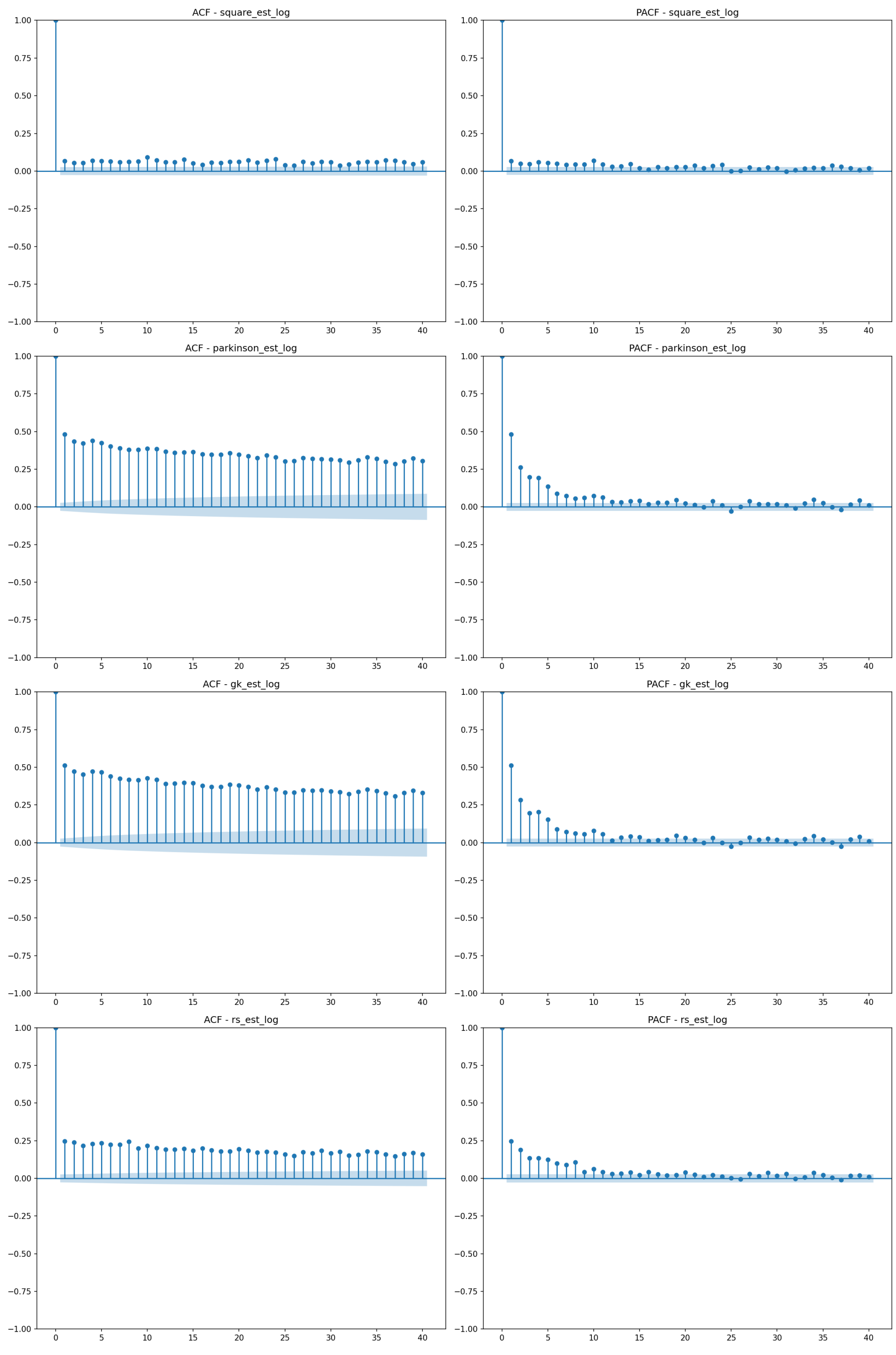

Figure 1: ACF and PACF for All Volatility Estimators

(1260, ‘name’)	square_est_log	parkinson_est_log	gk_est_log	rs_est_log

HAR Model Predictions vs True RV

The following plots compare the predicted volatility from each estimator against the true realized volatility.

HAR Model: Predictions vs True RV (Window=252)

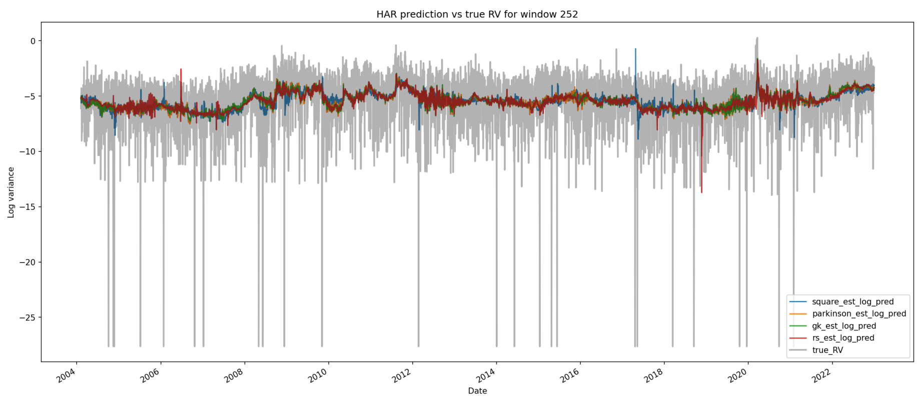

Figure 2: HAR Model: Predictions vs True RV (Window=252)

HAR Model: Predictions vs True RV (Window=504)

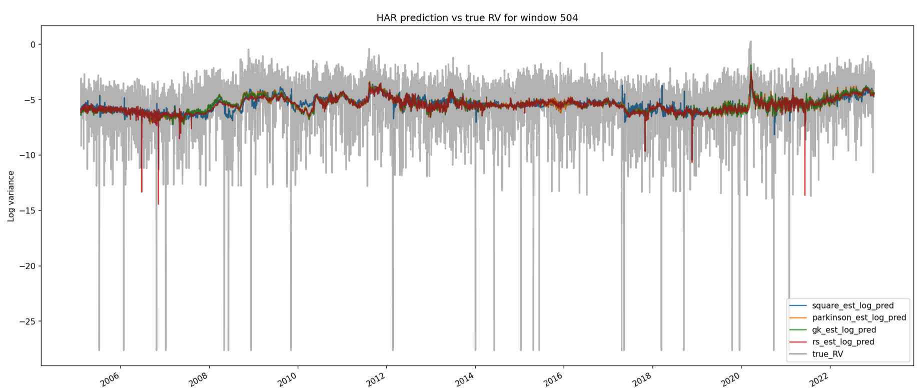

Figure 3: HAR Model: Predictions vs True RV (Window=504)

HAR Model: Predictions vs True RV (Window=756)

HAR Model: Predictions vs True RV (Window=1008)

HAR Model: Predictions vs True RV (Window=1260)

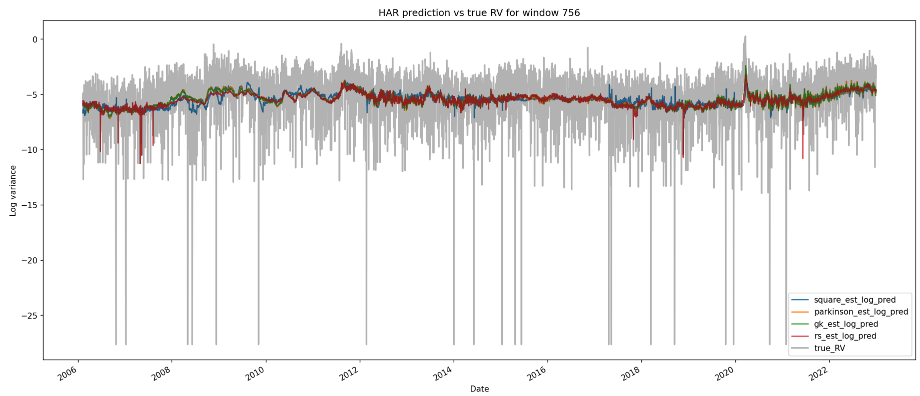

Figure 4: HAR Model: Predictions vs True RV (Window=756)

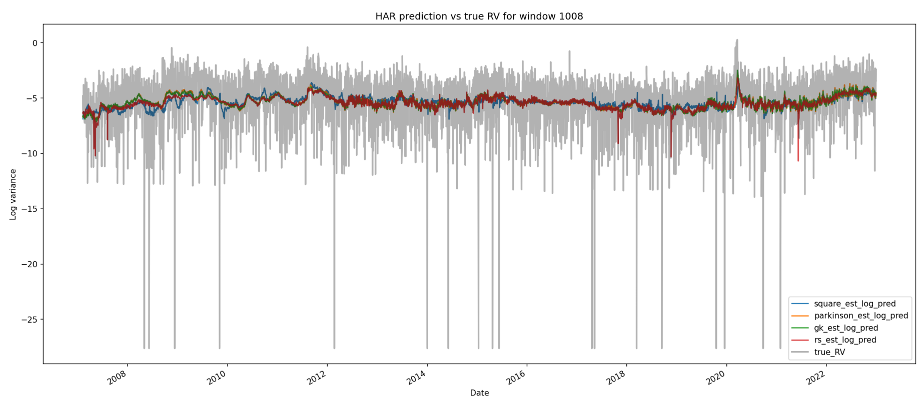

Figure 5: HAR Model: Predictions vs True RV (Window=1008)

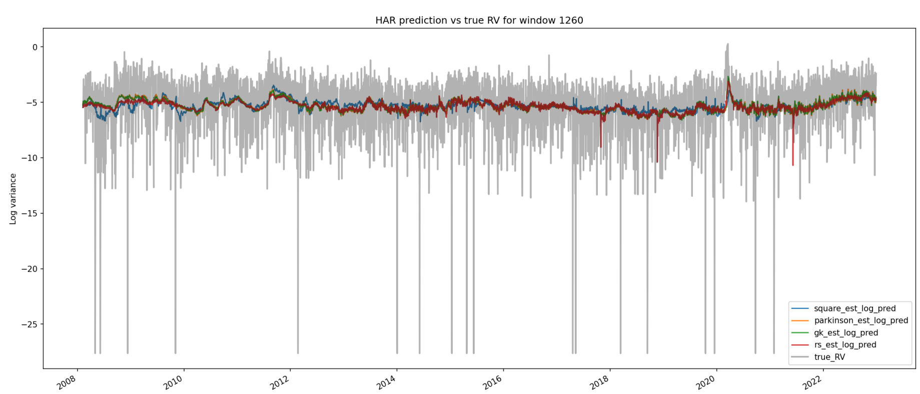

Figure 6: HAR Model: Predictions vs True RV (Window=1260)

Loss Metrics Over Time

QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude.

QLIKE Loss Over Time (Window=252)

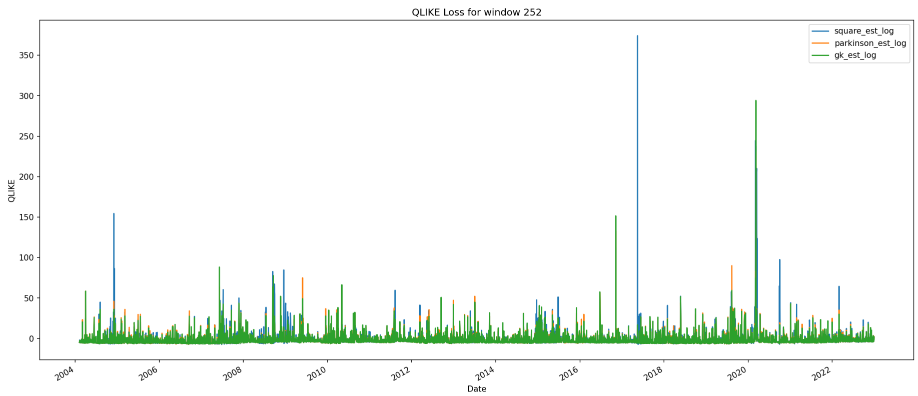

Figure 7: QLIKE Loss Over Time (Window=252)

QLIKE Loss Over Time (Window=504)

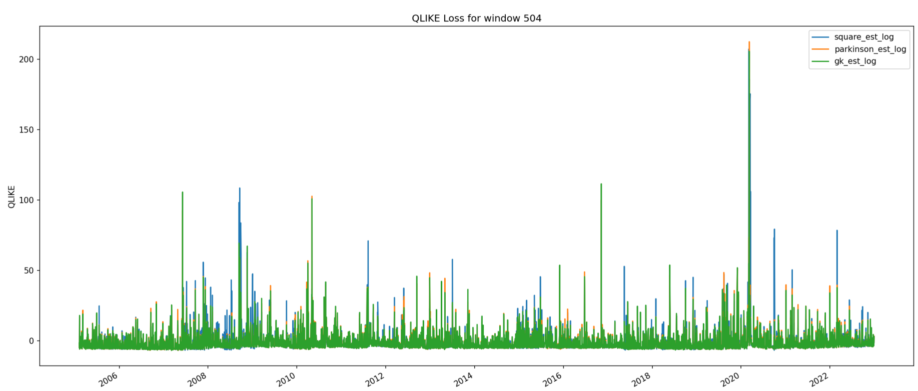

Figure 8: QLIKE Loss Over Time (Window=504)

QLIKE Loss Over Time (Window=756)

QLIKE Loss Over Time (Window=1008)

QLIKE Loss Over Time (Window=1260)

MSPE Loss Over Time (Window=252)

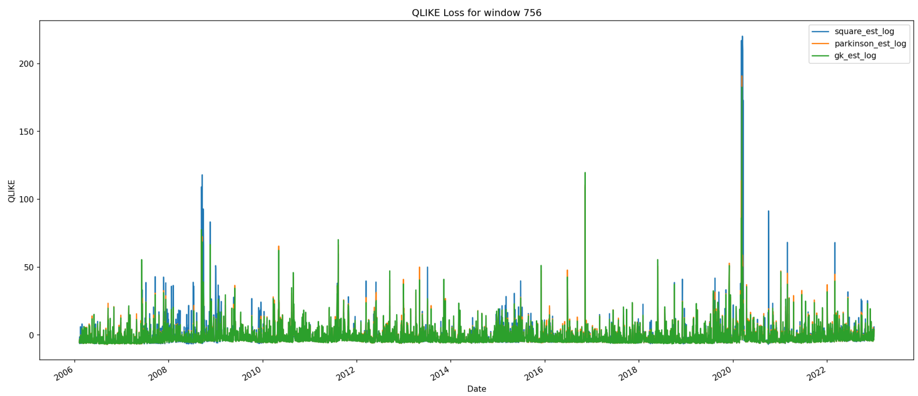

Figure 9: QLIKE Loss Over Time (Window=756)

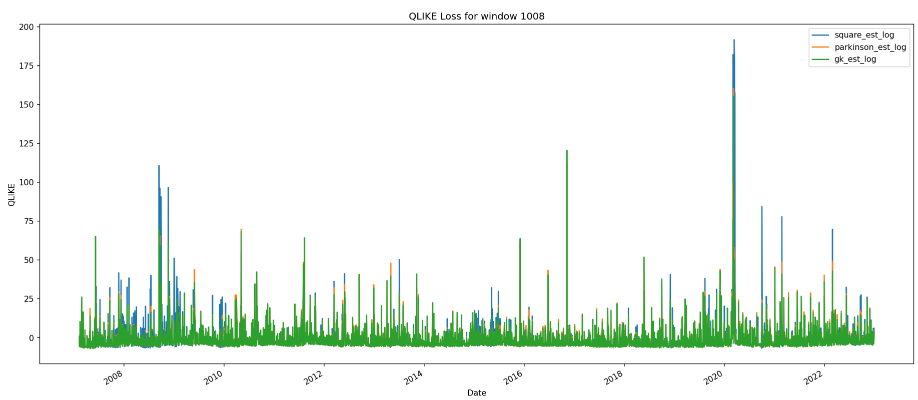

Figure 10: QLIKE Loss Over Time (Window=1008)

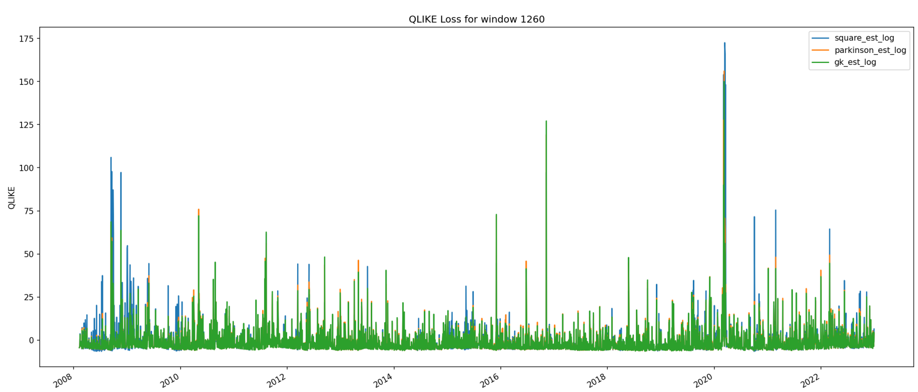

Figure 11: QLIKE Loss Over Time (Window=1260)

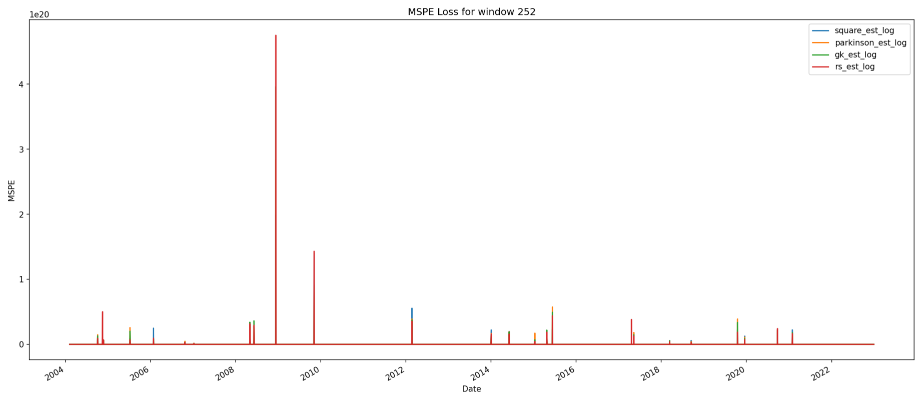

Figure 12: MSPE Loss Over Time (Window=252)

MSPE Loss Over Time (Window=504)

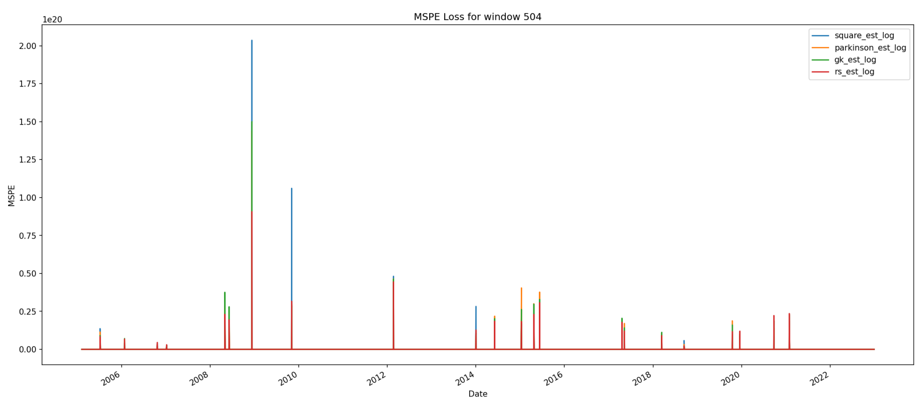

Figure 13: MSPE Loss Over Time (Window=504)

MSPE Loss Over Time (Window=756)

MSPE Loss Over Time (Window=1008)

MSPE Loss Over Time (Window=1260)

Ensemble Model Results

Ensemble Weights

The ensemble model combines predictions from multiple estimators using inverse QLIKE weighting. Below are the weights assigned to each estimator for different window sizes.

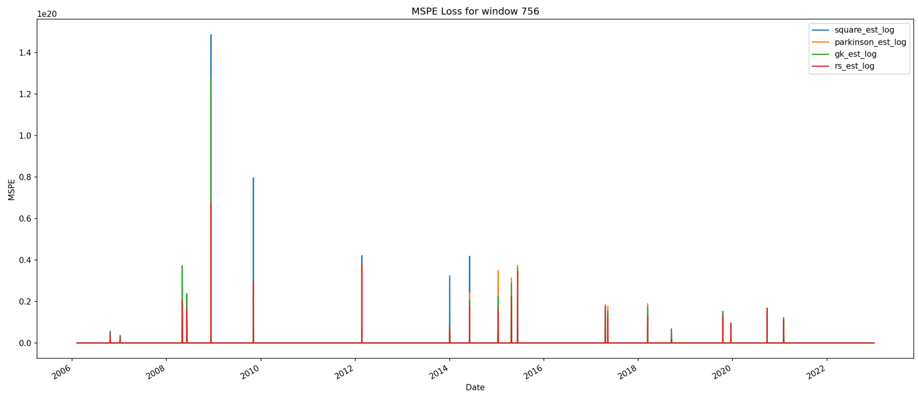

Figure 14: MSPE Loss Over Time (Window=756)

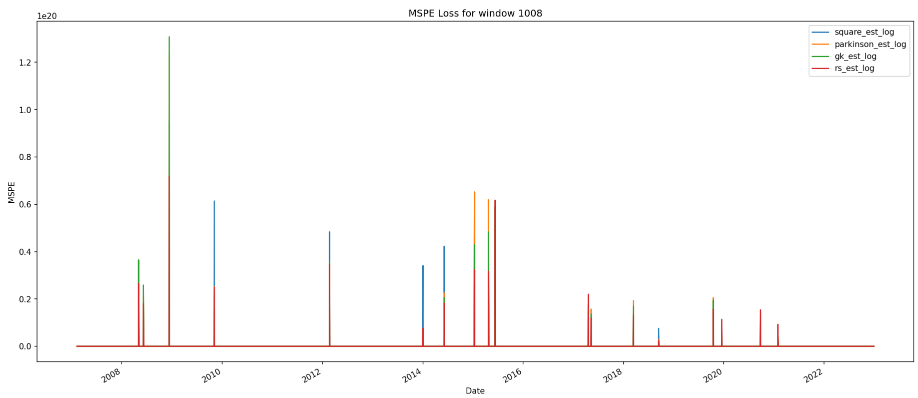

Figure 15: MSPE Loss Over Time (Window=1008)

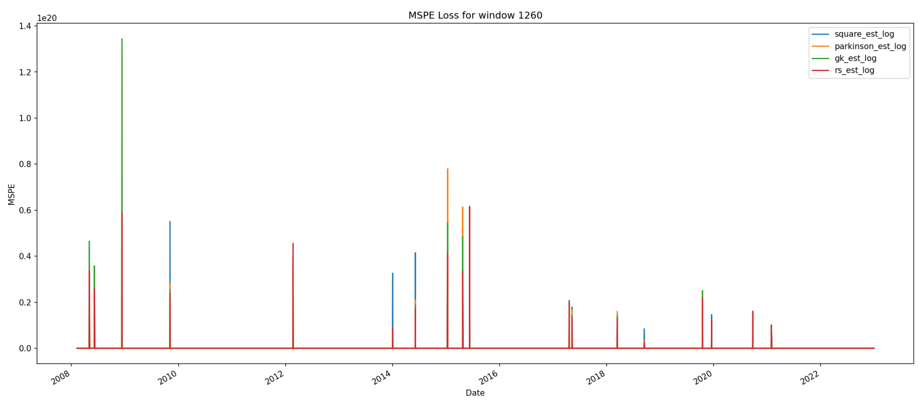

Figure 16: MSPE Loss Over Time (Window=1260)

Table 4: Ensemble Model Weights by Window

Window	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
252	0.25	0.25	0.25	0.25
504	0.333	0.333	0.333	0
756	0.25	0.25	0.25	0.25
1008	0.25	0.25	0.25	0.25
1260	0.25	0.25	0.25	0.25

Ensemble Performance Summary

Table 5: Ensemble Model Performance Metrics

	QLIKE_mean	QLIKE_std	MSPE_mean	MSPE_std
(252, 0)	-1.296	8.109	1.73417e+17	5.23932e+18
(504, 0)	-1.26	7.615	1.21603e+17	2.87704e+18
(756, 0)	-1.113	7.987	1.05214e+17	2.21583e+18
(1008, 0)	-1.051	7.824	1.25336e+17	2.44018e+18
(1260, 0)	-1.048	7.819	1.41157e+17	2.56842e+18

Table 6: Ensemble Model Ljung-Box Test Results (Part 1/2)

	lb_stat_10	lb_p_10	lb_stat_20	lb_p_20
(252, 0)	8.273	0.602	14.126	0.824
(252, 1)	8.273	0.602	14.126	0.824
(504, 0)	6.612	0.762	12.008	0.916
(504, 1)	6.612	0.762	12.008	0.916
(756, 0)	9.037	0.529	12.573	0.895
(756, 1)	9.037	0.529	12.573	0.895
(1008, 0)	10.321	0.413	13.816	0.84
(1008, 1)	10.321	0.413	13.816	0.84
(1260, 0)	10.375	0.408	14.274	0.816
(1260, 1)	10.375	0.408	14.274	0.816

Table 6: Ensemble Model Ljung-Box Test Results (Part 2/2)

	white_noise_flag	lb_lags_used	n_obs	name
(252, 0)	True	10	4759	
(252, 1)	True	20	4759	
(504, 0)	True	10	4507	
(504, 1)	True	20	4507	

	white_noise_flag	lb_lags_used	n_obs	name
(756, 0)	True	10	4255	
(756, 1)	True	20	4255	
(1008, 0)	True	10	4003	
(1008, 1)	True	20	4003	
(1260, 0)	True	10	3751	
(1260, 1)	True	20	3751	

Ensemble Predictions and Loss Metrics

Ensemble Model: Predictions vs True RV (Window=252)

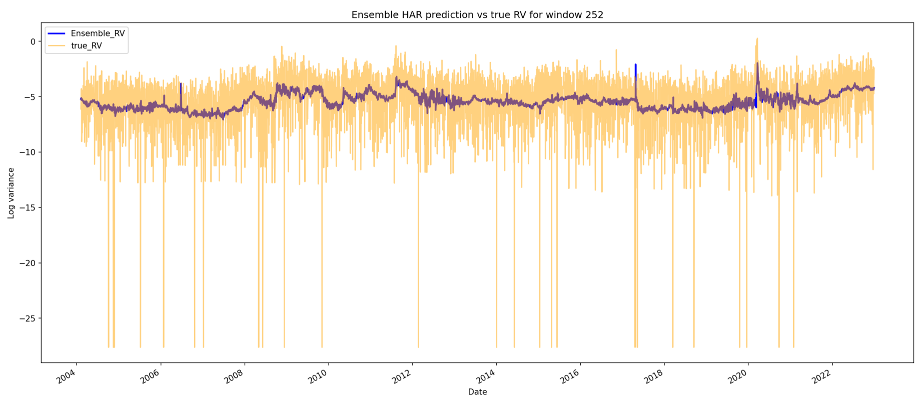

Figure 17: Ensemble Model: Predictions vs True RV (Window=252)

Ensemble Model: Predictions vs True RV (Window=504)

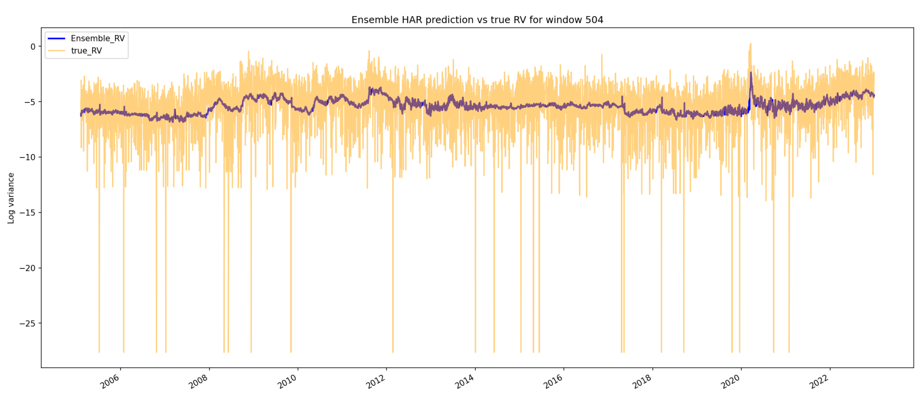

Figure 18: Ensemble Model: Predictions vs True RV (Window=504)

Ensemble Model: Predictions vs True RV (Window=756)

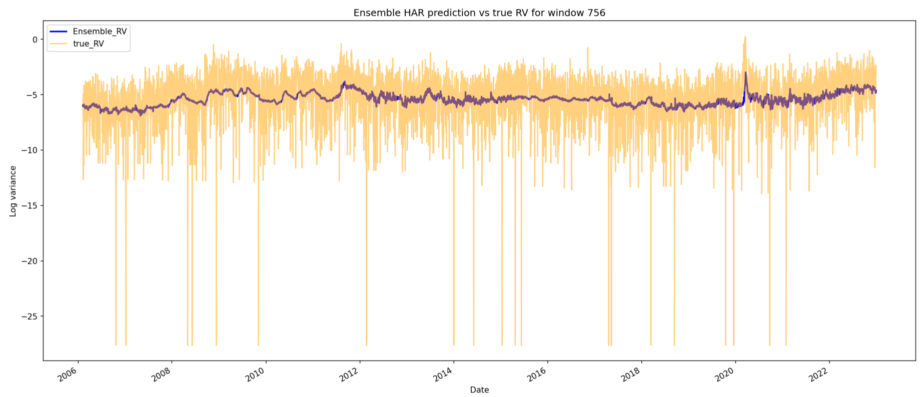

Figure 19: Ensemble Model: Predictions vs True RV (Window=756)

Ensemble Model: Predictions vs True RV (Window=1008)

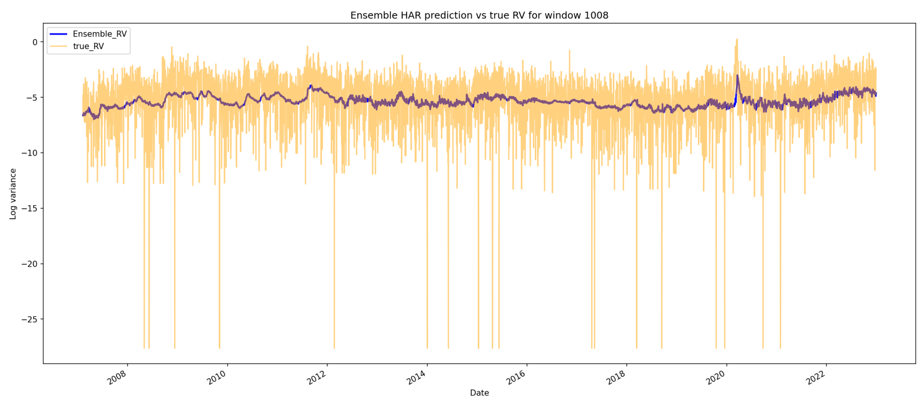

Figure 20: Ensemble Model: Predictions vs True RV (Window=1008)

Ensemble Model: Predictions vs True RV (Window=1260)

Ensemble QLIKE Loss (Window=252)

Ensemble QLIKE Loss (Window=504)

Ensemble QLIKE Loss (Window=756)

Ensemble QLIKE Loss (Window=1008)

Ensemble QLIKE Loss (Window=1260)

Ensemble MSPE Loss (Window=252)

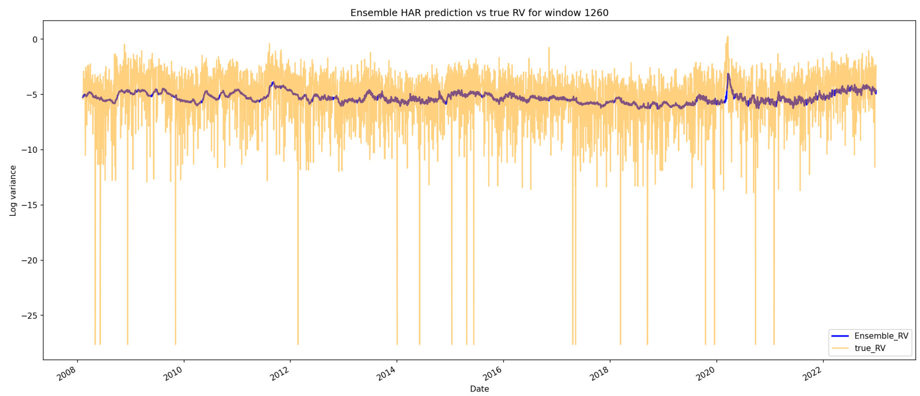

Figure 21: Ensemble Model: Predictions vs True RV (Window=1260)

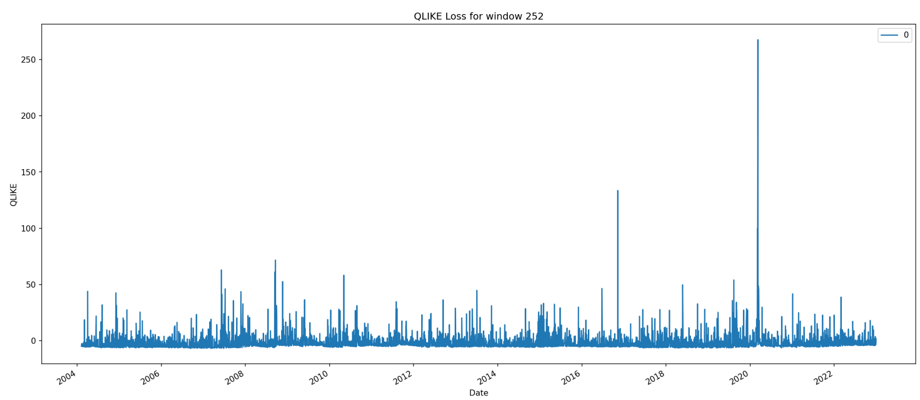

Figure 22: Ensemble QLIKE Loss (Window=252)

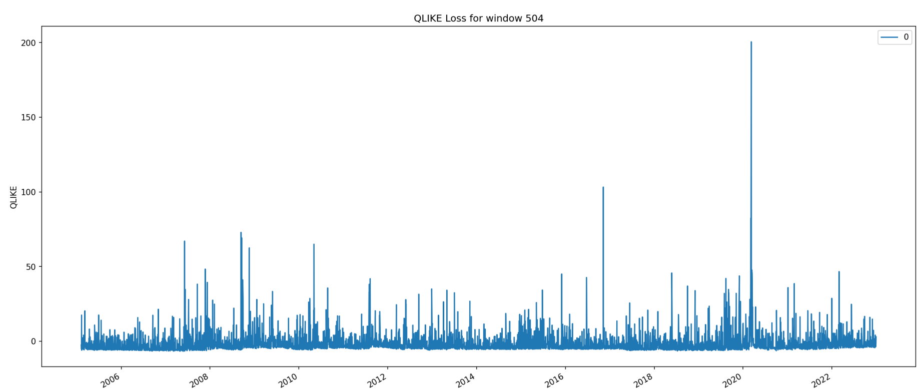

Figure 23: Ensemble QLIKE Loss (Window=504)

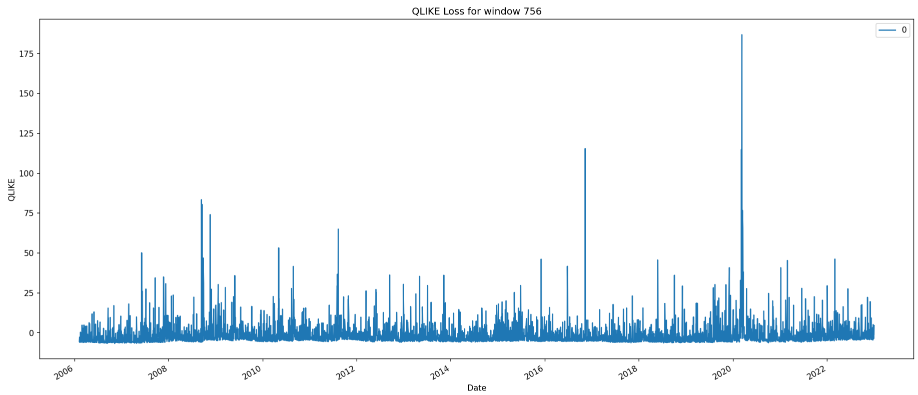

Figure 24: Ensemble QLIKE Loss (Window=756)

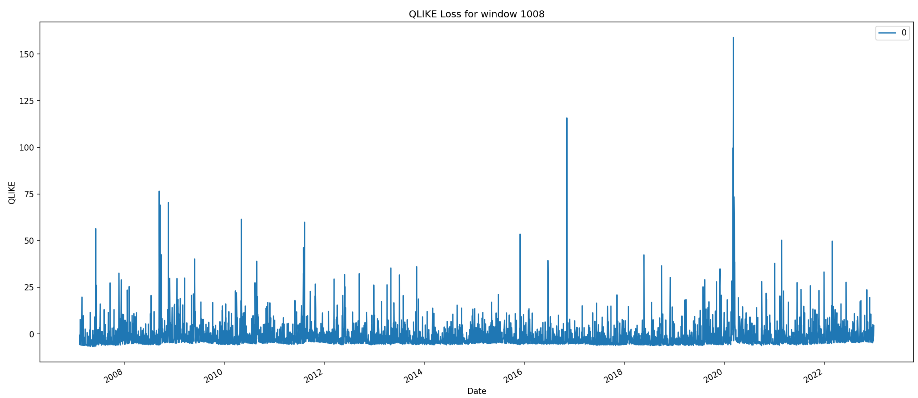

Figure 25: Ensemble QLIKE Loss (Window=1008)

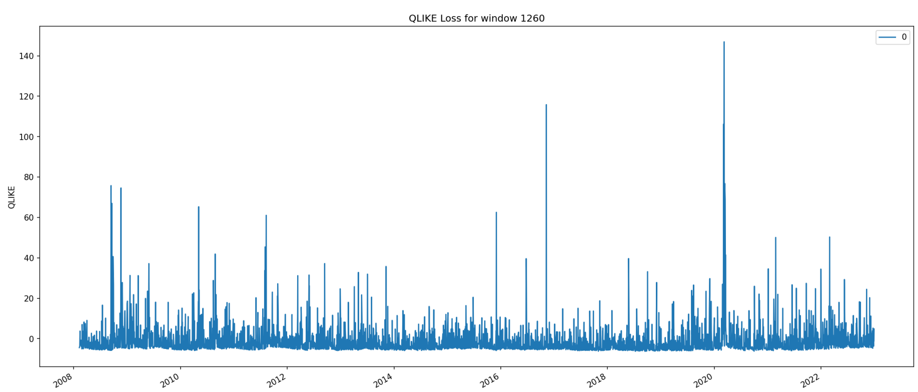

Figure 26: Ensemble QLIKE Loss (Window=1260)

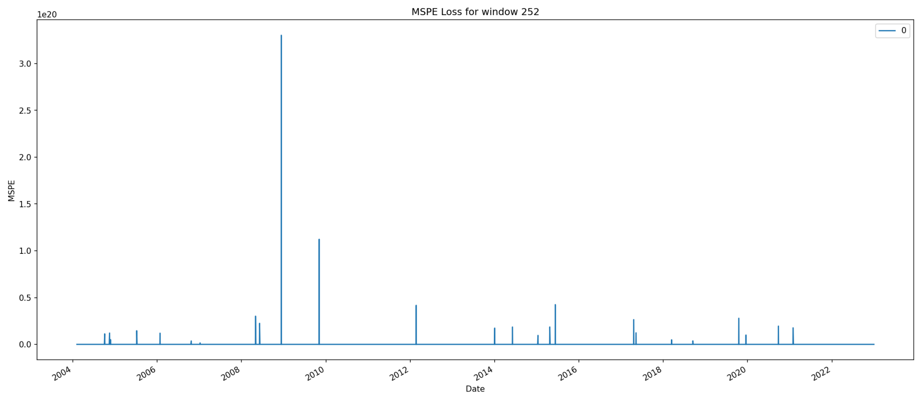

Figure 27: Ensemble MSPE Loss (Window=252)

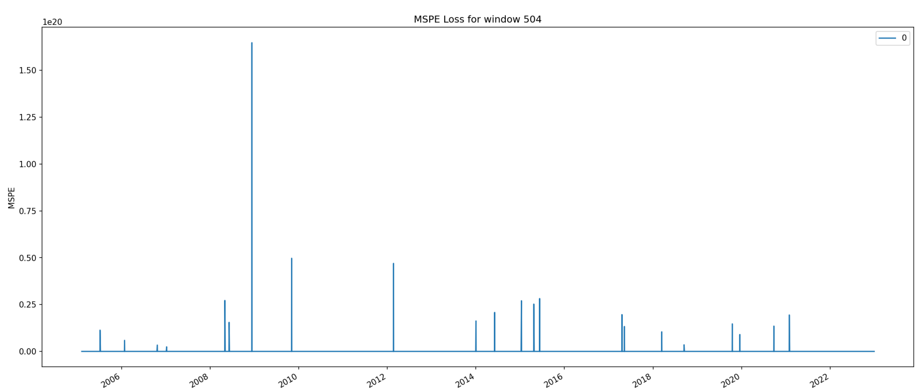

Figure 28: Ensemble MSPE Loss (Window=504)

Ensemble MSPE Loss (Window=504)

Ensemble MSPE Loss (Window=756)

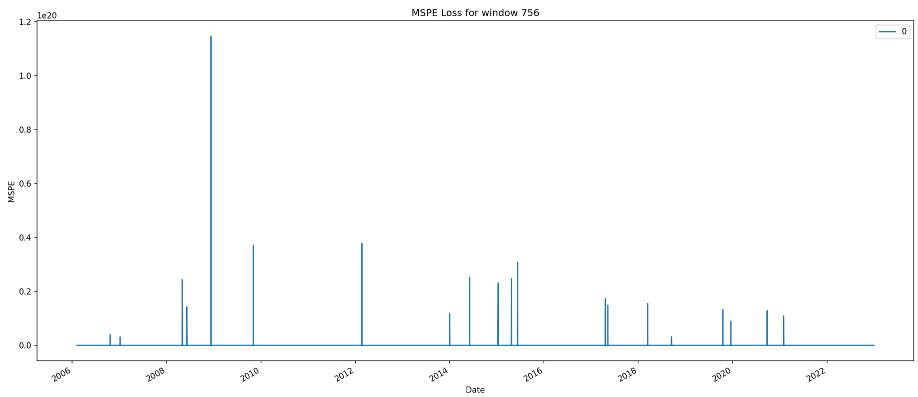

Figure 29: Ensemble MSPE Loss (Window=756)

Ensemble MSPE Loss (Window=1008)

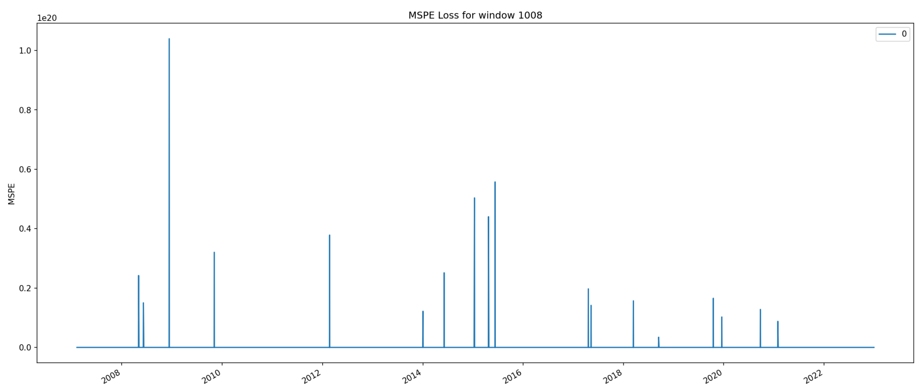

Figure 30: Ensemble MSPE Loss (Window=1008)

Ensemble MSPE Loss (Window=1260)

Diebold-Mariano Test

The Diebold-Mariano (DM) test evaluates whether there is a statistically significant difference in predictive accuracy between two models. Here we compare Windows 504 and 756.

DM Test Results: Window 504 vs Window 756

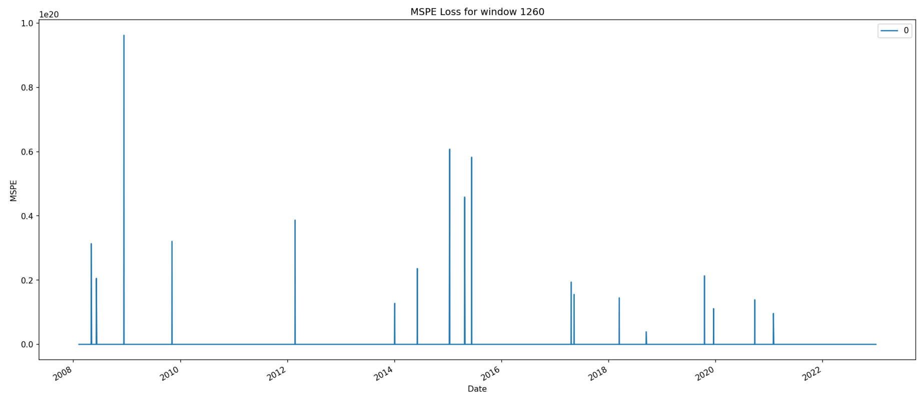

Figure 31: Ensemble MSPE Loss (Window=1260)

Metric	Value
DM Statistic	-1.4633
P-value	0.1434
Better Model	None (No significant difference)
Significant?	0.0000
Alpha	0.0500
Observations	4255.0000

Interpretation: With p-value = 0.1434 > 0.05, we fail to reject the null hypothesis of equal predictive accuracy. This indicates no statistically significant difference between Window 504 and 756. Either window may be used, and selection can be based on secondary metrics or practical considerations.

HAR-X Model Results

Exogenous Variables

The HAR-X model extends the HAR model by incorporating exogenous variables: - **UST10Y**: 10-Year US Treasury Yield - **HYOAS**: High Yield Option-Adjusted Spread - **TermSpread__10Y__2Y**: Term Spread (10Y - 2Y) - **VIX**: CBOE Volatility Index - **Breakeven10Y**: 10-Year Breakeven Inflation Rate

All exogenous variables were standardized using expanding window standardization to prevent look-ahead bias.

Table 7: Diagnostic Tests for Exogenous Variables (After Differencing) (Part 1/3)

Estimator	ADF stat	ADF p	ADF pass (p)	KPSS stat
UST10Y	-3.431	0.01	True	1.567
HYOAS	-3.298	0.015	True	0.882
TermSpread_10Y_2Y	-17.061	0	True	0.048
VIX	-5.957	0	True	0.463
Breakeven10Y	-4.528	0	True	0.823

Table 7: Diagnostic Tests for Exogenous Variables (After Differencing) (Part 2/3)

Estimator	KPSS p	KPSS pass (p>)	LB p @10	LB p @20
UST10Y	0.01	False	0	0
HYOAS	0.01	False	0	0
TermSpread_10Y_2Y	0.1	True	0	0
VIX	0.05	False	0	0
Breakeven10Y	0.01	False	0	0

Table 7: Diagnostic Tests for Exogenous Variables (After Differencing) (Part 3/3)

Estimator	White noise (LB)	Stationary (ADF KPSS)
UST10Y	0	0
HYOAS	0	0
TermSpread_10Y_2Y	0	1
VIX	0	0
Breakeven10Y	0	0

HARX Model Performance

The HAR-X model performance across different rolling window sizes is presented below.

Table 8: HAR-X Model Performance Summary

	QLIKE_mean	QLIKE_std	MSPE_mean	MSPE_std
(252, ‘square_est_log’)	-0.831	14.471	2.9745e+17	7.70254e+18
(252, ‘parkinson_est_log’)	-1.052	9.402	3.05783e+17	8.70434e+18
(252, ‘gk_est_log’)	-1.002	9.675	3.13035e+17	9.41853e+18
(252, ‘rs_est_log’)	-0.933	10.851	3.52866e+17	1.03554e+19
(504, ‘square_est_log’)	-1.249	7.798	2.24989e+17	5.43247e+18

	QLIKE_mean	QLIKE_std	MSPE_mean	MSPE_std
(504, 'parkinson_est_log')	-1.218	7.713	2.17787e+17	5.2861e+18
(504, 'gk_est_log')	-1.189	7.774	2.05416e+17	4.7914e+18
(504, 'rs_est_log')	-0.773	21.023	2.18173e+17	5.54433e+18
(756, 'square_est_log')	-1.308	6.913	2.49869e+17	7.24045e+18
(756, 'parkinson_est_log')	-1.277	6.95	2.28376e+17	6.59822e+18
(756, 'gk_est_log')	-1.268	6.944	2.19693e+17	6.15746e+18
(756, 'rs_est_log')	-0.994	14.202	2.27386e+17	6.76839e+18
(1008, 'square_est_log')	-1.181	6.962	2.34018e+17	5.99825e+18
(1008, 'parkinson_est_log')	-1.22	6.823	2.10392e+17	5.0361e+18
(1008, 'gk_est_log')	-1.214	6.806	2.06946e+17	4.99705e+18
(1008, 'rs_est_log')	-0.993	10.734	2.08147e+17	5.09185e+18
(1260, 'square_est_log')	-1.161	7.018	2.20667e+17	5.07197e+18
(1260, 'parkinson_est_log')	-1.201	6.922	2.12865e+17	4.66342e+18
(1260, 'gk_est_log')	-1.197	6.94	2.05688e+17	4.26511e+18
(1260, 'rs_est_log')	-1.031	9.766	1.94173e+17	4.05241e+18

Table 9: HAR-X Model Ljung-Box Test Results

	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(252, 'lb_stat_10')	24.70129355780[illegible]	[illegible]59723676090[illegible]	[illegible]900250192597[illegible]	[illegible]65999353625651
(252, 'lb_p_10')	0.005941386962[illegible]	[illegible]1822189[illegible]	[illegible]85139514590[illegible]	[illegible]96861744676368
(252, 'lb_stat_20')	31.745323100901[illegible]	[illegible]811115633951[illegible]	[illegible]4224296614[illegible]	[illegible]0753549933427
(252, 'lb_p_20')	0.046088778580[illegible]	[illegible]946250807095[illegible]	[illegible]7613626832[illegible]	[illegible]87475144834253
(252, 'white_noise_flag')	False	True	True	True
(252, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(252, 'n_obs')	5256	5256	5256	5256
(252, 'name')	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(504, 'lb_stat_10')	10.76327007352[illegible]	[illegible]036286088217[illegible]	[illegible]454658599782[illegible]	[illegible]01725639313884
(504, 'lb_p_10')	0.376256512629[illegible]	[illegible]06091479194[illegible]	[illegible]94818054680[illegible]	[illegible]8901092376795
(504, 'lb_stat_20')	17.767715375532[illegible]	[illegible]716842695705[illegible]	[illegible]7575012953[illegible]	[illegible]5331486494261

	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(504, 'lb_p_20')	0.6027068025330[illegible]	[illegible]296409469134[illegible]	[illegible]109450569560[illegible]	[illegible]766799620039438
(504, 'white_noise_flag')	True	True	True	True
(504, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(504, 'n_obs')	5004	5004	5004	5004
(504, 'name')	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(756, 'lb_stat_10')	16.241550037010[illegible]	[illegible]966782419266[illegible]	[illegible]664470572931[illegible]	[illegible]96861802170436
(756, 'lb_p_10')	0.0929233901117[illegible]	[illegible]3976183466[illegible]	[illegible]5236465958[illegible]	[illegible]373479445041
(756, 'lb_stat_20')	22.267478748846[illegible]	[illegible]7319187664[illegible]	[illegible]528423016[illegible]	[illegible]8.935475521556253
(756, 'lb_p_20')	0.3261746921534[illegible]	[illegible]7408126760[illegible]	[illegible]64997560949[illegible]	[illegible]60236853958826
(756, 'white_noise_flag')	True	True	True	True
(756, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(756, 'n_obs')	4752	4752	4752	4752
(756, 'name')	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(1008, 'lb_stat_10')	20.44766402391[illegible]	[illegible]37991519541[illegible]	[illegible]2400469190[illegible]	[illegible]125791083441896
(1008, 'lb_p_10')	0.0252918133164[illegible]	[illegible]8723438154[illegible]	[illegible]6312291951[illegible]	[illegible]7542599308923
(1008, 'lb_stat_20')	25.876696692775[illegible]	[illegible]797984913[illegible]	[illegible]3035575227[illegible]	[illegible]523008163900503
(1008, 'lb_p_20')	0.1699229439074[illegible]	[illegible]4285036612[illegible]	[illegible]318350838[illegible]	[illegible]694386214413116
(1008, 'white_noise_flag')	False	True	True	True
(1008, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(1008, 'n_obs')	4500	4500	4500	4500
(1008, 'name')	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(1260, 'lb_stat_10')	18.115190446320[illegible]	[illegible]780352366743[illegible]	[illegible]41333006983[illegible]	[illegible]5941638356064
(1260, 'lb_p_10')	0.0530513522143[illegible]	[illegible]840697615[illegible]	[illegible]3200457293[illegible]	[illegible]84045793821869
(1260, 'lb_stat_20')	24.486503839167[illegible]	[illegible]778658083[illegible]	[illegible]185594163[illegible]	[illegible]729129809402796
(1260, 'lb_p_20')	0.2217862755932[illegible]	[illegible]6769954965[illegible]	[illegible]05355550827[illegible]	[illegible]199101705401047
(1260, 'white_noise_flag')	True	True	True	True

	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
(1260, 'lb_lags_used')	(10, 20)	(10, 20)	(10, 20)	(10, 20)
(1260, 'n_obs')	4248	4248	4248	4248
(1260, 'name')	square_est_log	parkinson_est_log	gk_est_log	rs_est_log

HAR-X Model Predictions vs True RV

The following plots compare the predicted volatility from each estimator with exogenous variables against the true realized volatility.

HAR-X Model: Predictions vs True RV (Window=252)

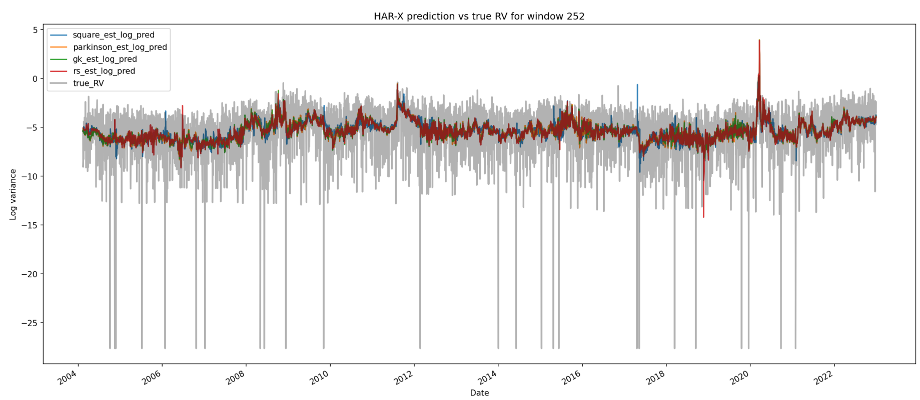

Figure 32: HAR-X Model: Predictions vs True RV (Window=252)

HAR-X Model: Predictions vs True RV (Window=504)

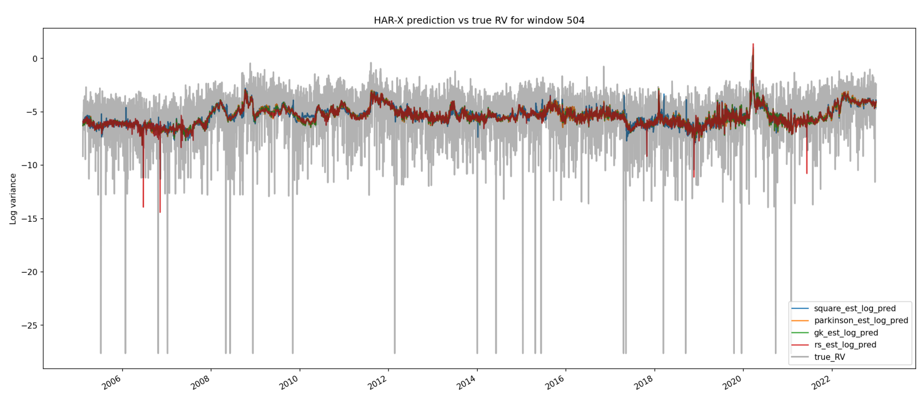

Figure 33: HAR-X Model: Predictions vs True RV (Window=504)

HAR-X Model: Predictions vs True RV (Window=756)

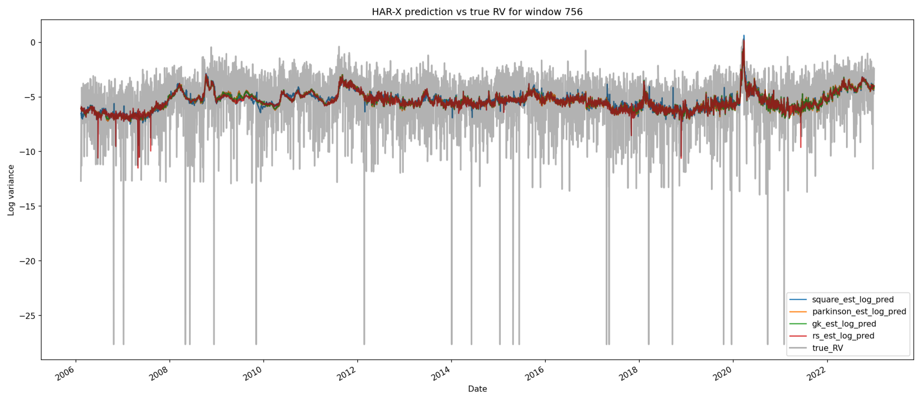

Figure 34: HAR-X Model: Predictions vs True RV (Window=756)

HAR-X Model: Predictions vs True RV (Window=1008)

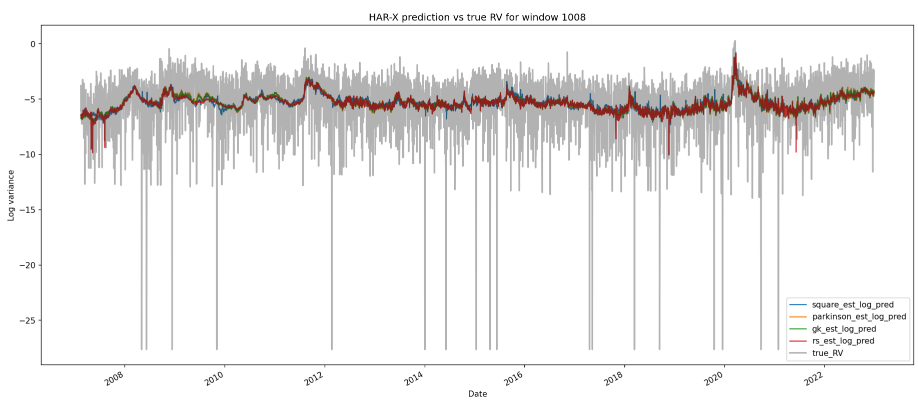

Figure 35: HAR-X Model: Predictions vs True RV (Window=1008)

HAR-X Model: Predictions vs True RV (Window=1260)

Loss Metrics Over Time

QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude for the HAR-X model with exogenous variables.

HAR-X QLIKE Loss Over Time (Window=252)

HAR-X QLIKE Loss Over Time (Window=504)

HAR-X QLIKE Loss Over Time (Window=756)

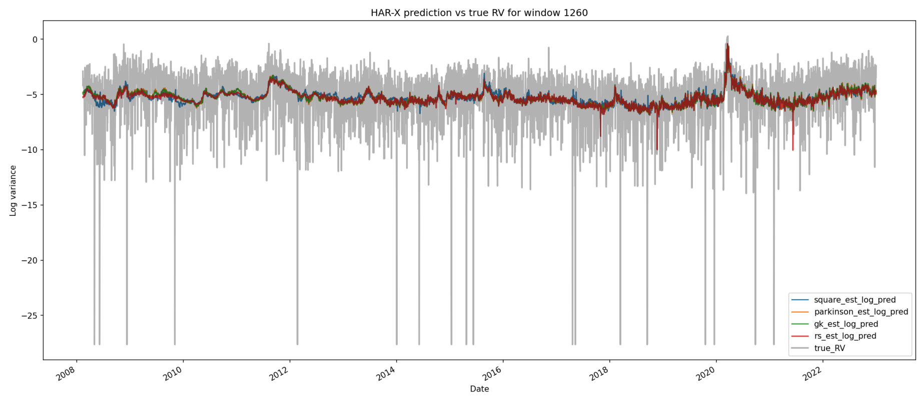

Figure 36: HAR-X Model: Predictions vs True RV (Window=1260)

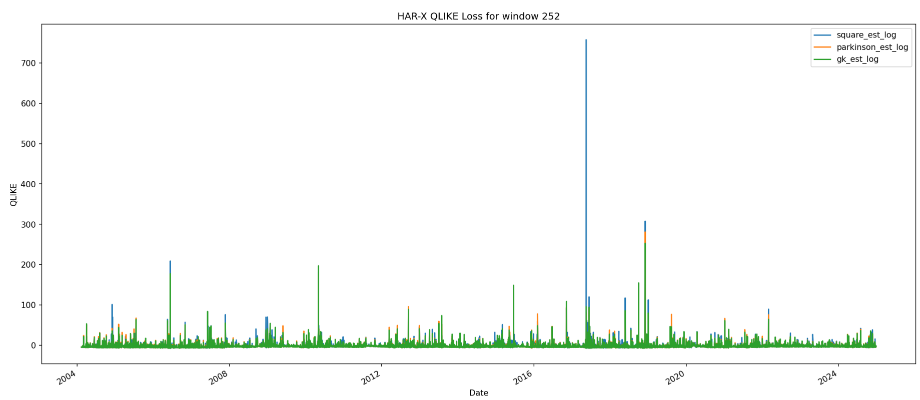

Figure 37: HAR-X QLIKE Loss Over Time (Window=252)

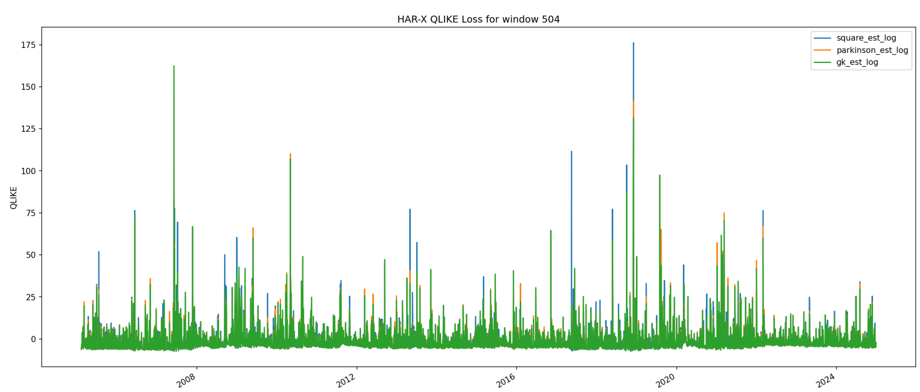

Figure 38: HAR-X QLIKE Loss Over Time (Window=504)

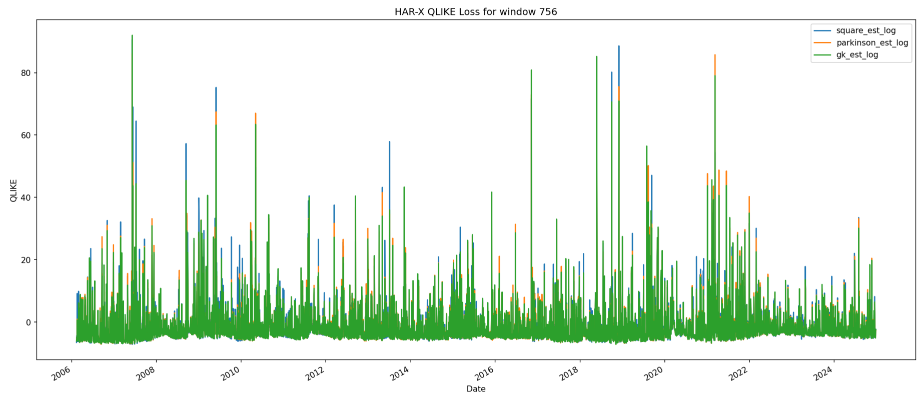

Figure 39: HAR-X QLIKE Loss Over Time (Window=756)

HAR-X QLIKE Loss Over Time (Window=1008)

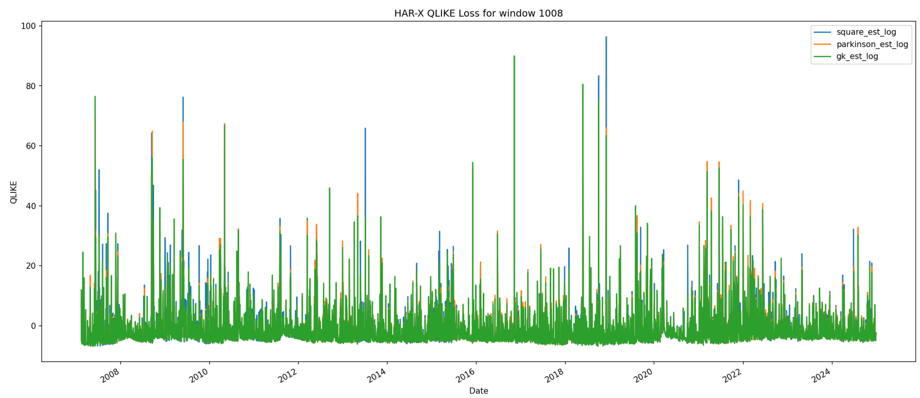

Figure 40: HAR-X QLIKE Loss Over Time (Window=1008)

HAR-X QLIKE Loss Over Time (Window=1260)

HAR-X MSPE Loss Over Time (Window=252)

HAR-X MSPE Loss Over Time (Window=504)

HAR-X MSPE Loss Over Time (Window=756)

HAR-X MSPE Loss Over Time (Window=1008)

HAR-X MSPE Loss Over Time (Window=1260)

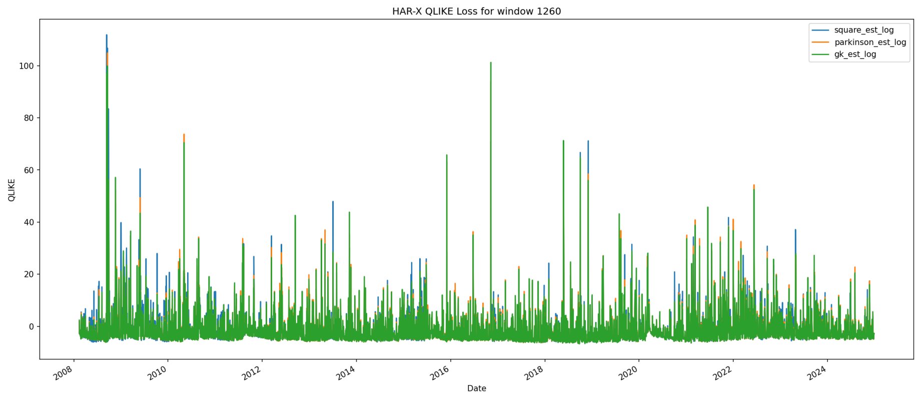

Figure 41: HAR-X QLIKE Loss Over Time (Window=1260)

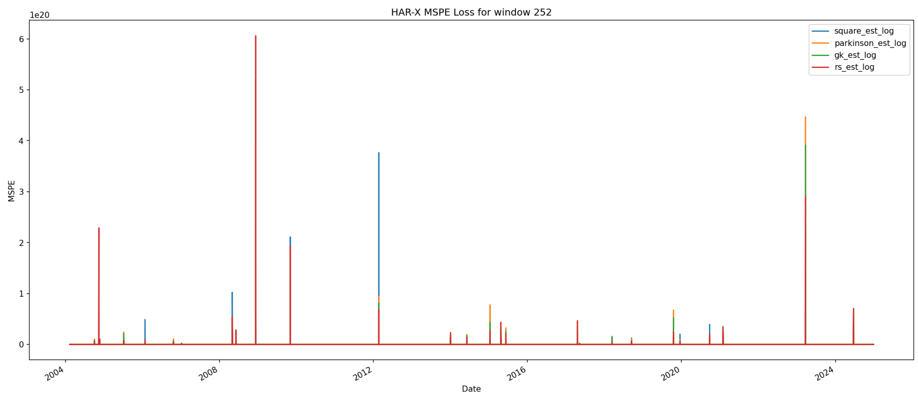

Figure 42: HAR-X MSPE Loss Over Time (Window=252)

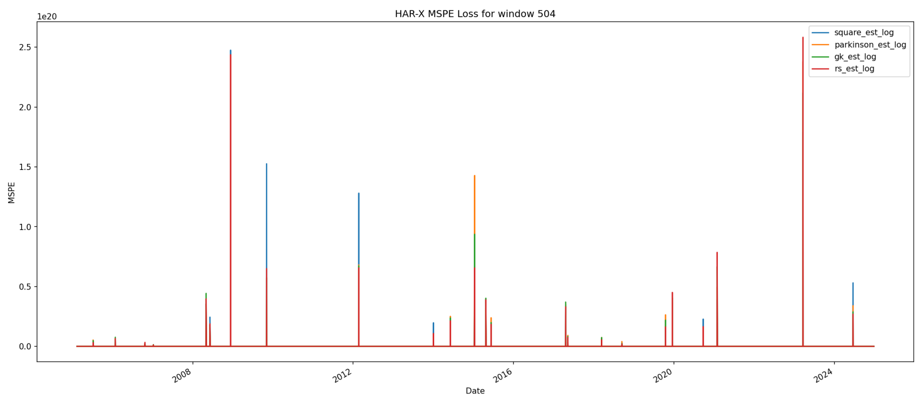

Figure 43: HAR-X MSPE Loss Over Time (Window=504)

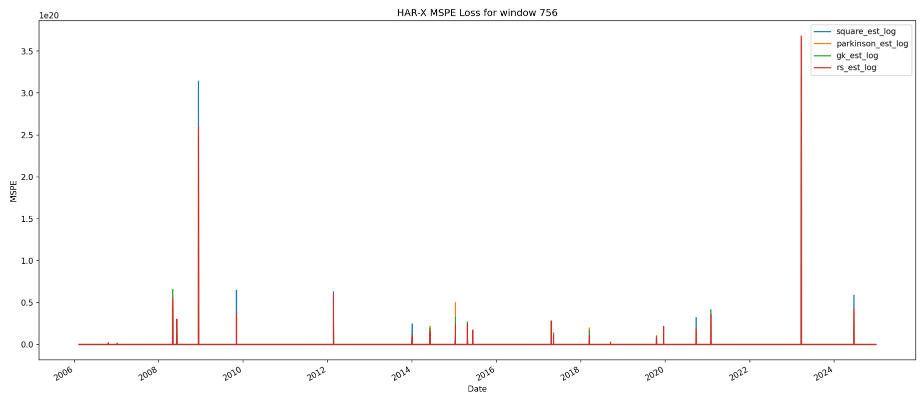

Figure 44: HAR-X MSPE Loss Over Time (Window=756)

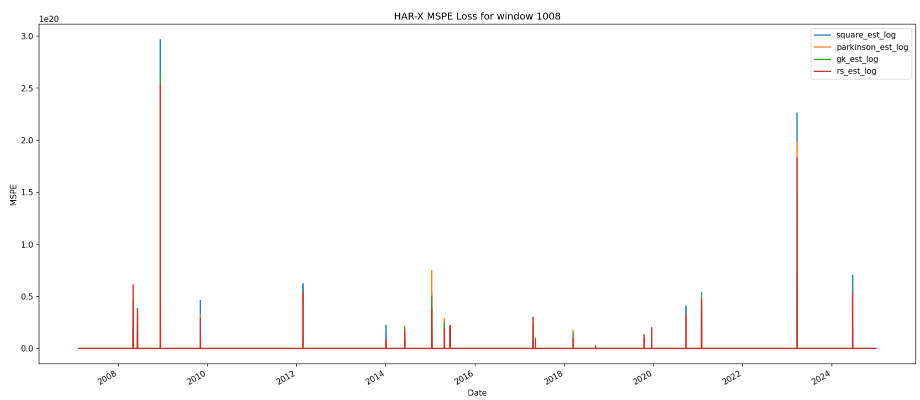

Figure 45: HAR-X MSPE Loss Over Time (Window=1008)

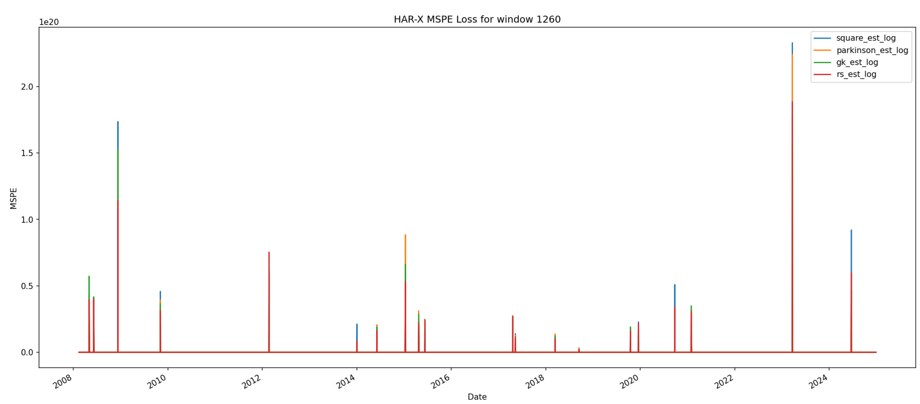

Figure 46: HAR-X MSPE Loss Over Time (Window=1260)

HAR-X Ensemble Model Results

Ensemble Weights

The HAR-X ensemble model combines predictions from multiple estimators using inverse QLIKE weighting. Below are the weights assigned to each estimator for different window sizes.

Table 10: HAR-X Ensemble Model Weights by Window

Window	square_est_log	parkinson_est_log	gk_est_log	rs_est_log
252	0.25	0.25	0.25	0.25
504	0.25	0.25	0.25	0.25
756	0.25	0.25	0.25	0.25
1008	0.25	0.25	0.25	0.25
1260	0.25	0.25	0.25	0.25

HAR-X Ensemble Performance Summary

Table 11: HAR-X Ensemble Model Performance Metrics

	QLIKE_mean	QLIKE_std	MSPE_mean	MSPE_std
(252, 0)	-1.167	9.249	2.48679e+17	7.04661e+18
(504, 0)	-1.268	7.505	1.80769e+17	4.25835e+18
(756, 0)	-1.272	6.987	1.63743e+17	4.44177e+18
(1008, 0)	-1.179	6.949	1.79972e+17	4.61601e+18
(1260, 0)	-1.175	7.064	1.62983e+17	3.19451e+18

Table 12: HAR-X Ensemble Model Ljung-Box Test Results (Part 1/2)

	lb_stat_10	lb_p_10	lb_stat_20	lb_p_20
(252, 0)	7.828	0.646	13.448	0.857
(252, 1)	7.828	0.646	13.448	0.857
(504, 0)	5.97	0.818	12.122	0.912
(504, 1)	5.97	0.818	12.122	0.912
(756, 0)	9.95	0.445	14.529	0.803
(756, 1)	9.95	0.445	14.529	0.803
(1008, 0)	13.334	0.206	17.386	0.628
(1008, 1)	13.334	0.206	17.386	0.628
(1260, 0)	11.796	0.299	16.221	0.703
(1260, 1)	11.796	0.299	16.221	0.703

Table 12: HAR-X Ensemble Model Ljung-Box Test Results (Part 2/2)

	white_noise_flag	lb_lags_used	n_obs	name
(252, 0)	True	10	4756	
(252, 1)	True	20	4756	
(504, 0)	True	10	4504	
(504, 1)	True	20	4504	
(756, 0)	True	10	4252	
(756, 1)	True	20	4252	
(1008, 0)	True	10	4000	
(1008, 1)	True	20	4000	
(1260, 0)	True	10	3748	
(1260, 1)	True	20	3748	

HAR-X Ensemble Predictions and Loss Metrics

HAR-X Ensemble Model: Predictions vs True RV (Window=252)

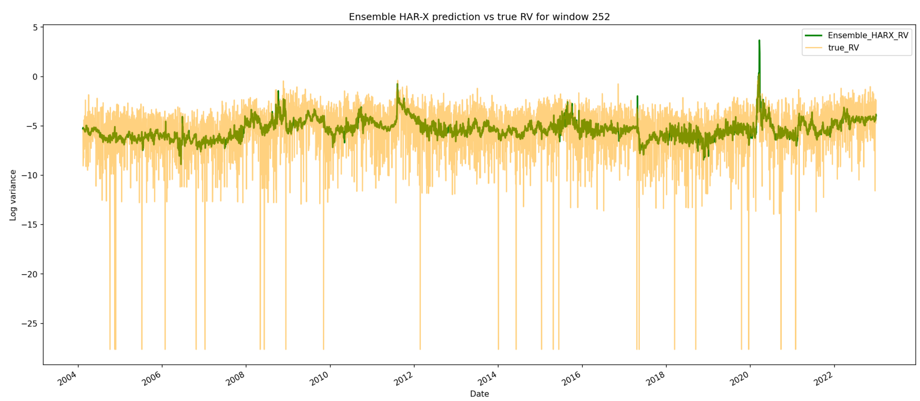

Figure 47: HAR-X Ensemble Model: Predictions vs True RV (Window=252)

HAR-X Ensemble Model: Predictions vs True RV (Window=504)

HAR-X Ensemble Model: Predictions vs True RV (Window=756)

HAR-X Ensemble Model: Predictions vs True RV (Window=1008)

HAR-X Ensemble Model: Predictions vs True RV (Window=1260)

HAR-X Ensemble QLIKE Loss (Window=252)

HAR-X Ensemble QLIKE Loss (Window=504)

HAR-X Ensemble QLIKE Loss (Window=756)

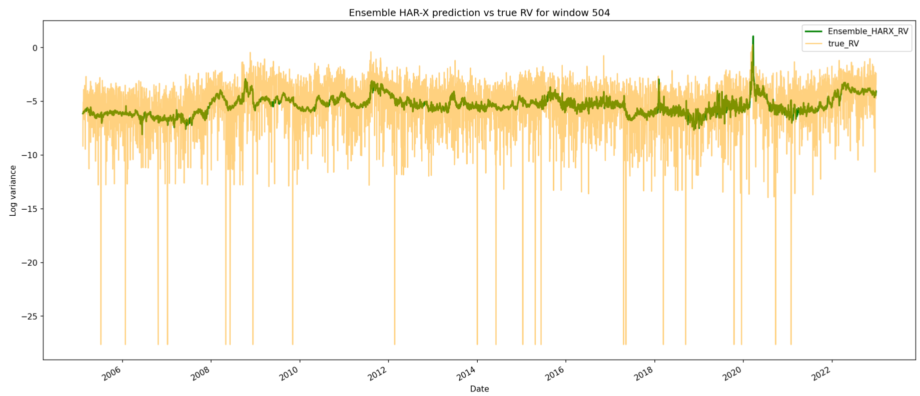

Figure 48: HAR-X Ensemble Model: Predictions vs True RV (Window=504)

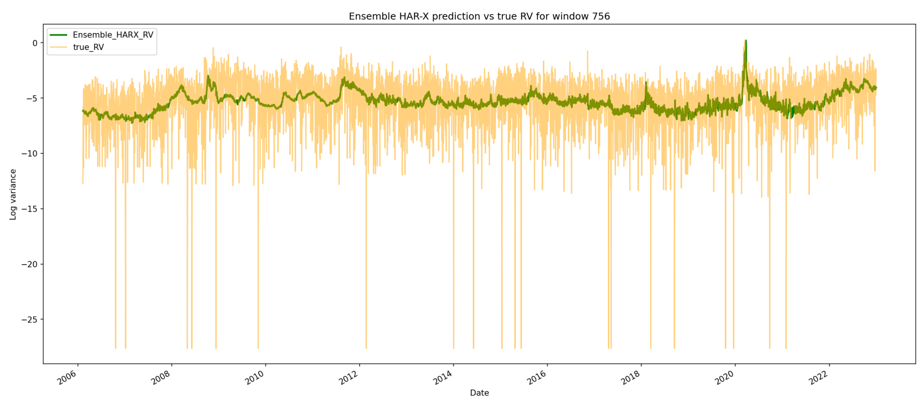

Figure 49: HAR-X Ensemble Model: Predictions vs True RV (Window=756)

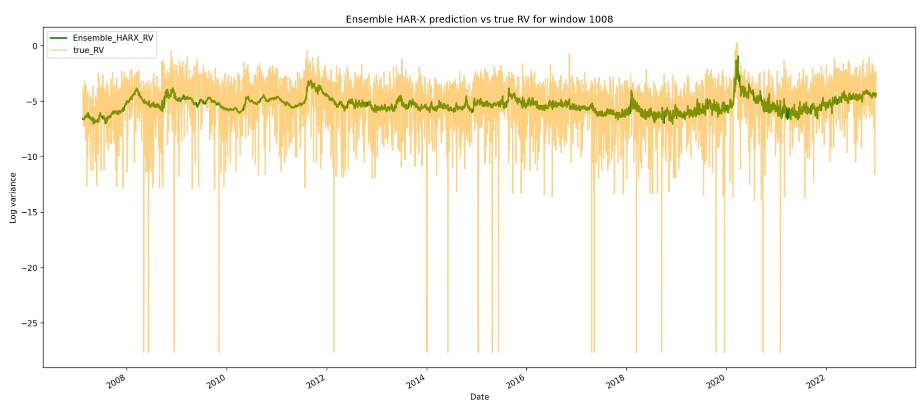

Figure 50: HAR-X Ensemble Model: Predictions vs True RV (Window=1008)

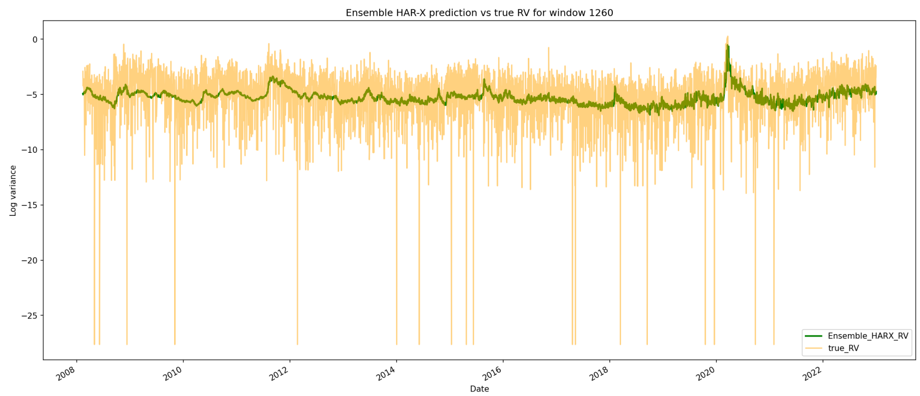

Figure 51: HAR-X Ensemble Model: Predictions vs True RV (Window=1260)

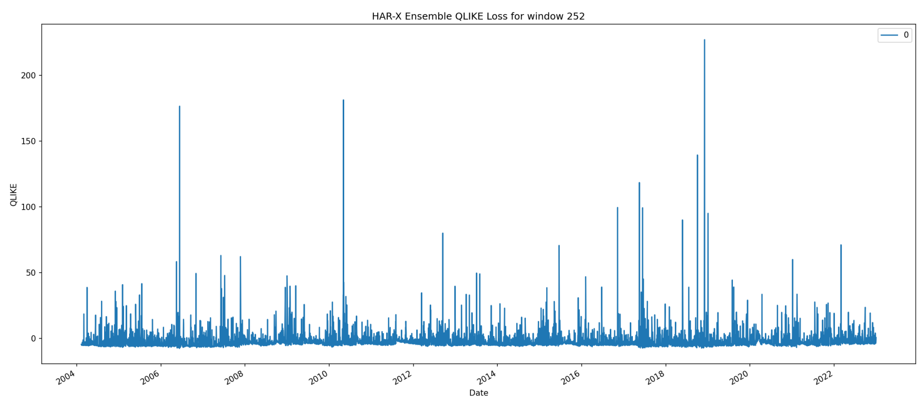

Figure 52: HAR-X Ensemble QLIKE Loss (Window=252)

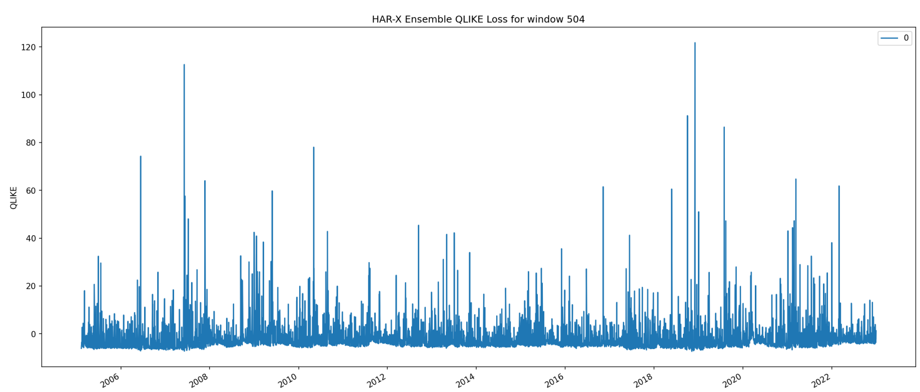

Figure 53: HAR-X Ensemble QLIKE Loss (Window=504)

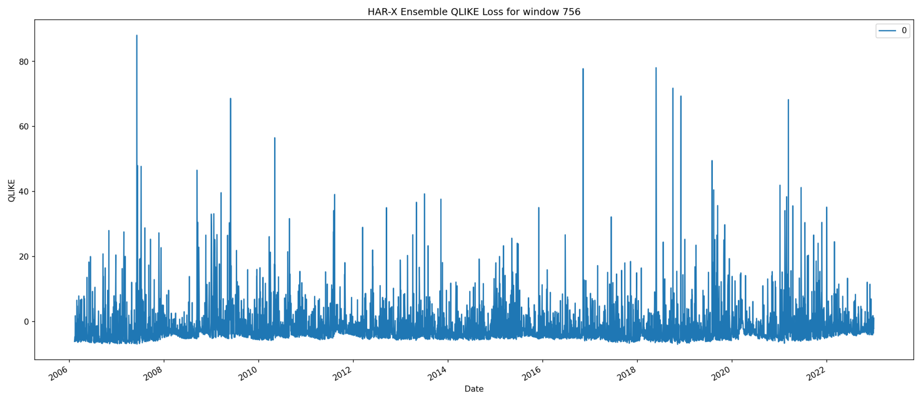

Figure 54: HAR-X Ensemble QLIKE Loss (Window=756)

HAR-X Ensemble QLIKE Loss (Window=1008)

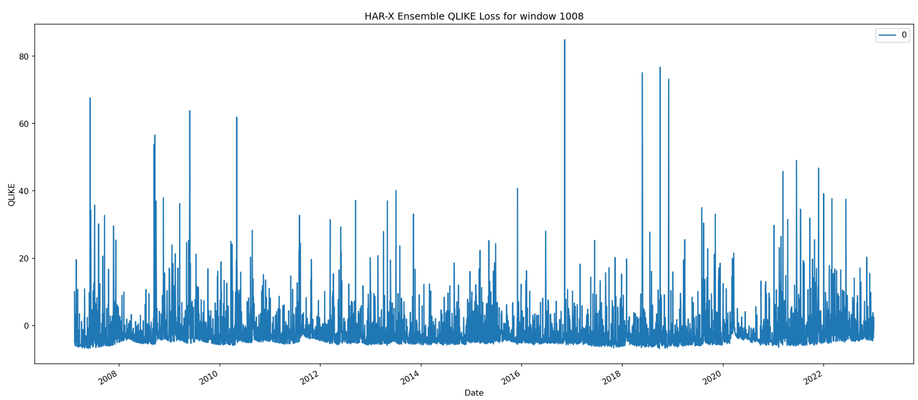

Figure 55: HAR-X Ensemble QLIKE Loss (Window=1008)

HAR-X Ensemble QLIKE Loss (Window=1260)

HAR-X Ensemble MSPE Loss (Window=252)

HAR-X Ensemble MSPE Loss (Window=504)

HAR-X Ensemble MSPE Loss (Window=756)

HAR-X Ensemble MSPE Loss (Window=1008)

HAR-X Ensemble MSPE Loss (Window=1260)

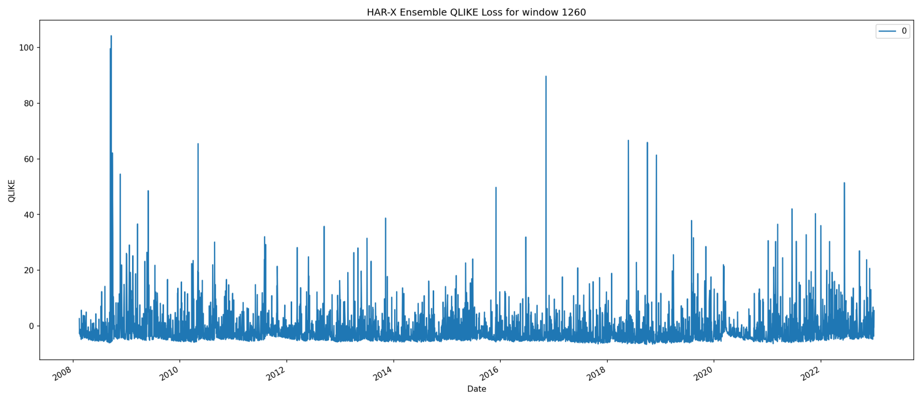

Figure 56: HAR-X Ensemble QLIKE Loss (Window=1260)

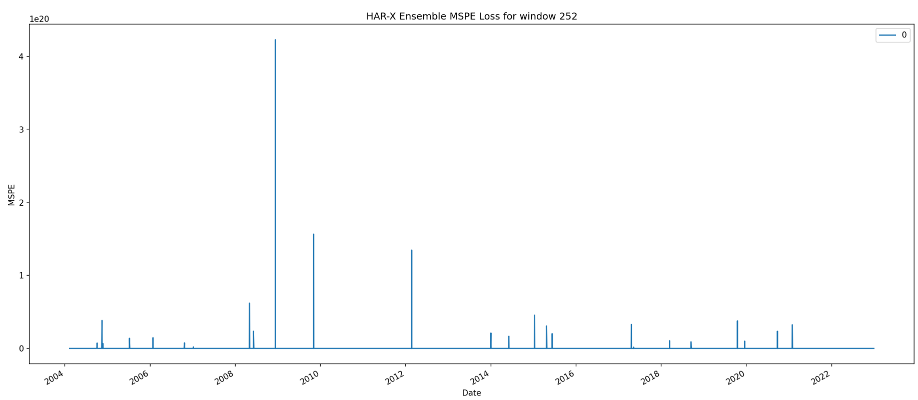

Figure 57: HAR-X Ensemble MSPE Loss (Window=252)

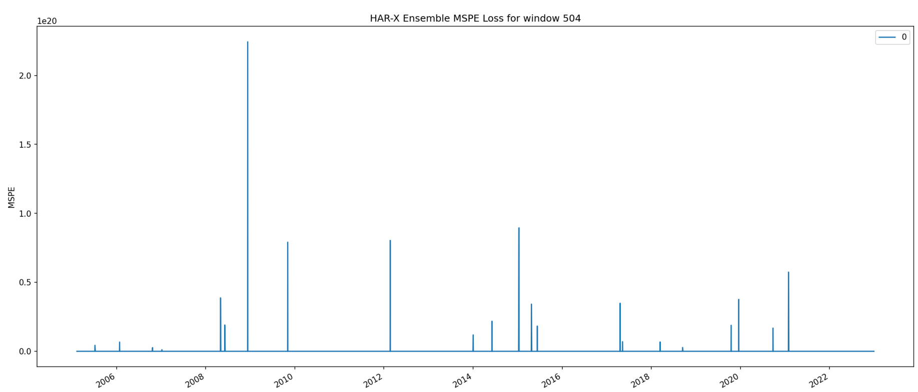

Figure 58: HAR-X Ensemble MSPE Loss (Window=504)

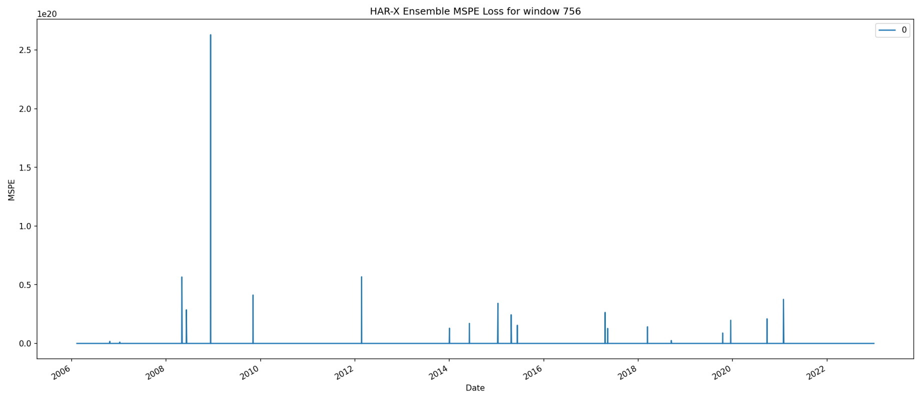

Figure 59: HAR-X Ensemble MSPE Loss (Window=756)

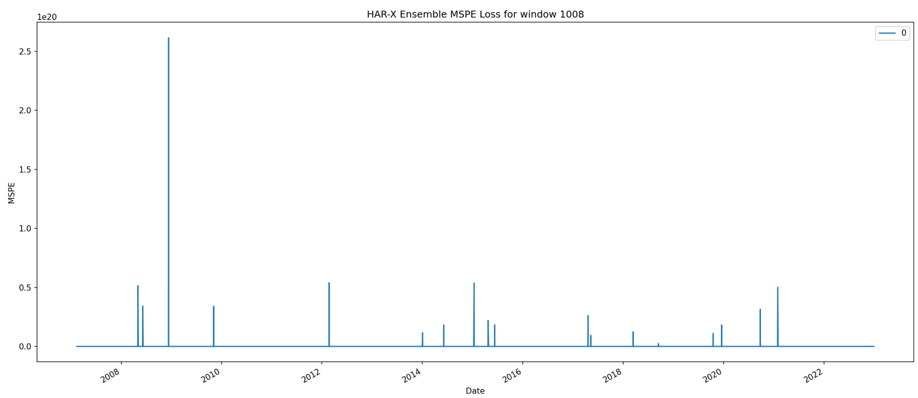

Figure 60: HAR-X Ensemble MSPE Loss (Window=1008)

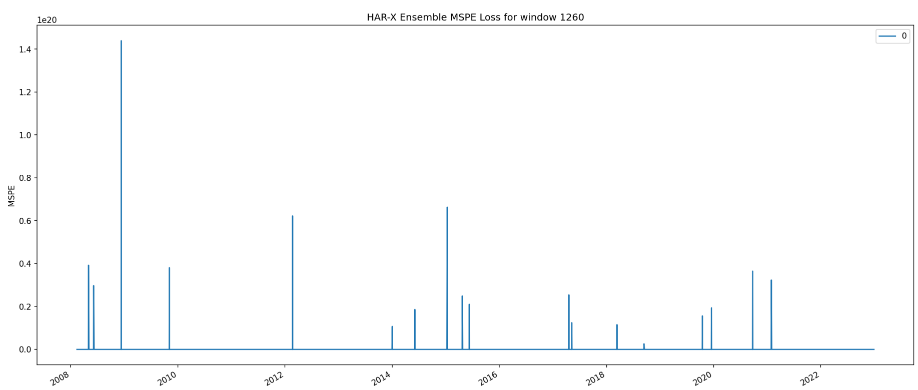

Figure 61: HAR-X Ensemble MSPE Loss (Window=1260)

HAR-X Diebold-Mariano Test

The Diebold-Mariano (DM) test evaluates whether there is a statistically significant difference in predictive accuracy between two HAR-X models. Here we compare Windows 504 and 756.

HAR-X DM Test Results: Window 504 vs Window 756

Metric	Value
DM Statistic	nan
P-value	nan
Better Model	None (No significant difference)
Significant?	0.0000
Alpha	0.0500
Observations	4752.0000

Interpretation: With p-value = nan, we fail to reject the null hypothesis of equal predictive accuracy. This indicates no statistically significant difference between Window 504 and 756 for the HAR-X model.

Report generation completed

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Machine Learning Models

This section evaluates five state-of-the-art machine learning algorithms for volatility forecasting: - **Random Forest (RF)**: Ensemble of decision trees - **Gradient Boosting Machine (GBM)**: Sequential tree learning with error correction - **XGBoost**: Optimized gradient boosting with GPU support - **LightGBM**: Fast gradient boosting with leaf-wise growth - **CatBoost**: Boosting optimized for categorical features

Each model is trained on a rolling window of historical volatility estimators and exogenous variables, then evaluated on out-of-sample predictions across three window sizes (252, 504, 756 days).

RF Model Performance

Random Forest (RF) is an ensemble method that builds multiple decision trees and averages their predictions. It's robust to outliers and naturally captures non-linear relationships in volatility forecasting without requiring feature scaling.

Table: RF Performance Summary Across Windows (Part 1/2)

	Window	QLIKE	QLIKE Std	MSPE
0	full	7.522	29.592	4391.26

Table: RF Performance Summary Across Windows (Part 2/2)

	MSPE Std	RMSE	Dir. Accuracy
0	55665	0.045	51.58%

RF Predictions vs True RV The following plots compare predicted volatility against true realized volatility for each rolling window.

RF: Predictions vs True RV (Window=full)

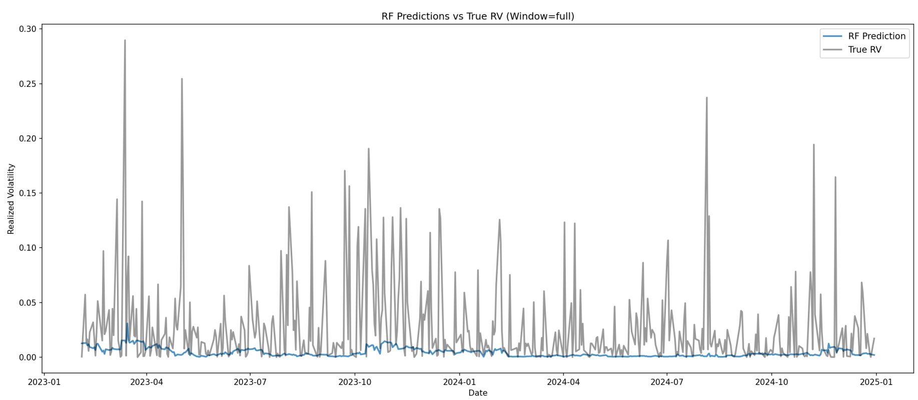

Figure 62: RF: Predictions vs True RV (Window=full)

RF Loss Metrics Over Time QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude for the ML model across different training periods.

RF QLIKE Loss Over Time (Window=full)

RF MSPE Loss Over Time (Window=full)

GBM Model Performance

Gradient Boosting Machine (GBM) sequentially builds decision trees, with each tree correcting the errors of previous ones. This incremental approach often provides superior performance for forecasting tasks by explicitly learning residual patterns.

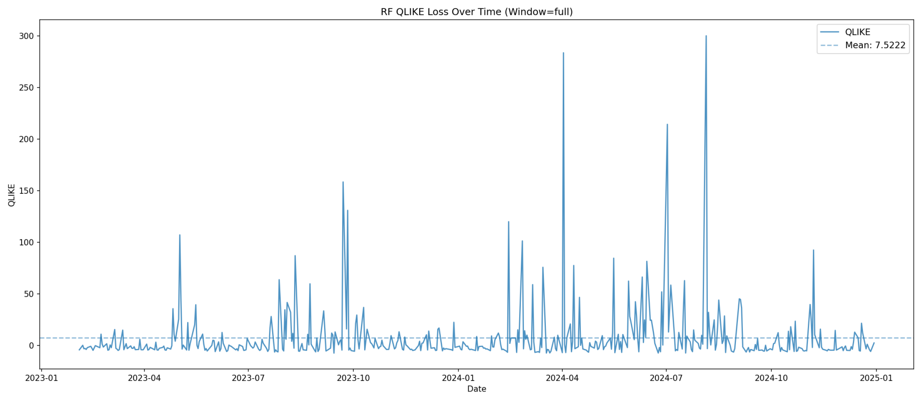

Figure 63: RF QLIKE Loss Over Time (Window=full)

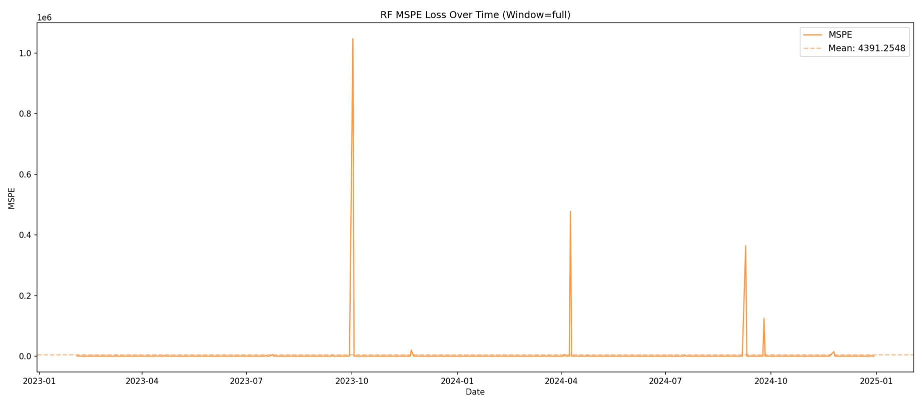

Figure 64: RF MSPE Loss Over Time (Window=full)

Table: GBM Performance Summary Across Windows (Part 1/2)

	Window	QLIKE	QLIKE Std	MSPE
0	full	24.459	146.674	9738.12

Table: GBM Performance Summary Across Windows (Part 2/2)

	MSPE Std	RMSE	Dir. Accuracy
0	159979	0.045	50.53%

GBM Predictions vs True RV The following plots compare predicted volatility against true realized volatility for each rolling window.

GBM: Predictions vs True RV (Window=full)

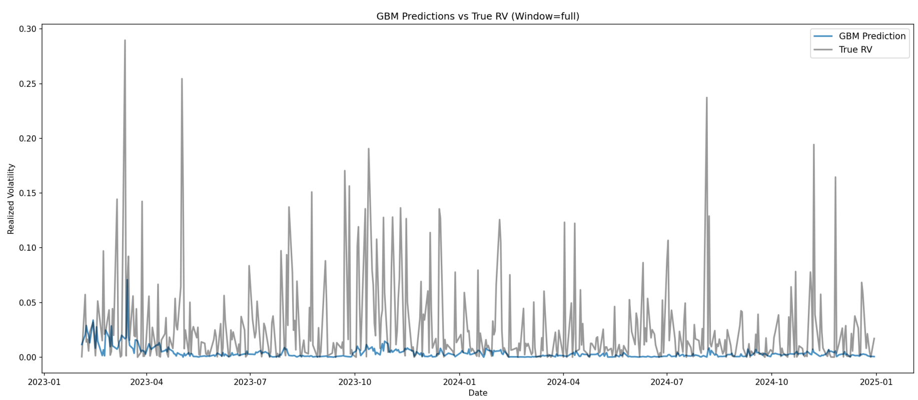

Figure 65: GBM: Predictions vs True RV (Window=full)

GBM Loss Metrics Over Time QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude for the ML model across different training periods.

GBM QLIKE Loss Over Time (Window=full)

GBM MSPE Loss Over Time (Window=full)

XGBOOST Model Performance

XGBoost (eXtreme Gradient Boosting) is an optimized gradient boosting implementation with GPU acceleration support. It includes regularization to prevent overfitting and has proven highly effective for time series forecasting competitions.

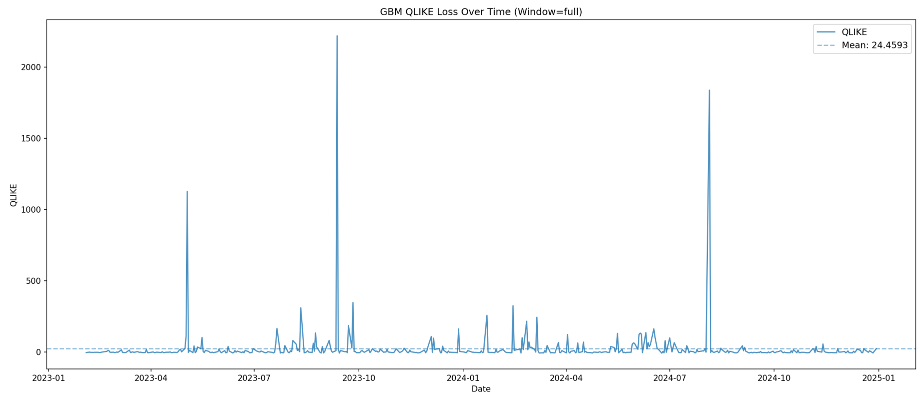

Figure 66: GBM QLIKE Loss Over Time (Window=full)

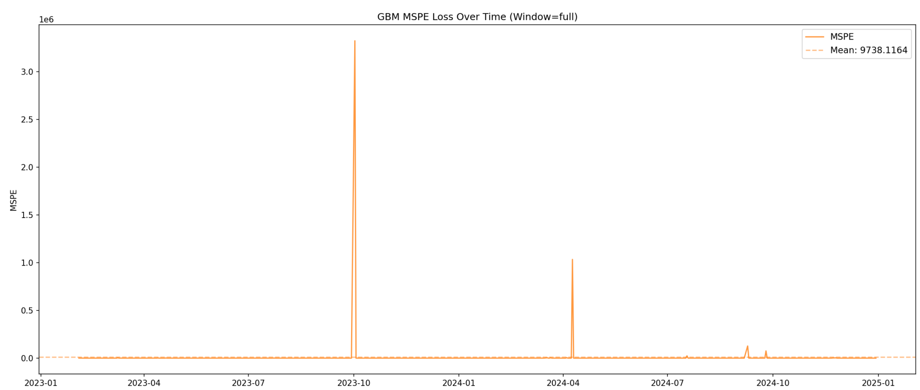

Figure 67: GBM MSPE Loss Over Time (Window=full)

Table: XGBOOST Performance Summary Across Windows (Part 1/2)

	Window	QLIKE	QLIKE Std	MSPE
0	full	14.502	64.309	7132.32

Table: XGBOOST Performance Summary Across Windows (Part 2/2)

	MSPE Std	RMSE	Dir. Accuracy
0	135627	0.046	45.05%

XGBOOST Predictions vs True RV The following plots compare predicted volatility against true realized volatility for each rolling window.

XGBOOST: Predictions vs True RV (Window=full)

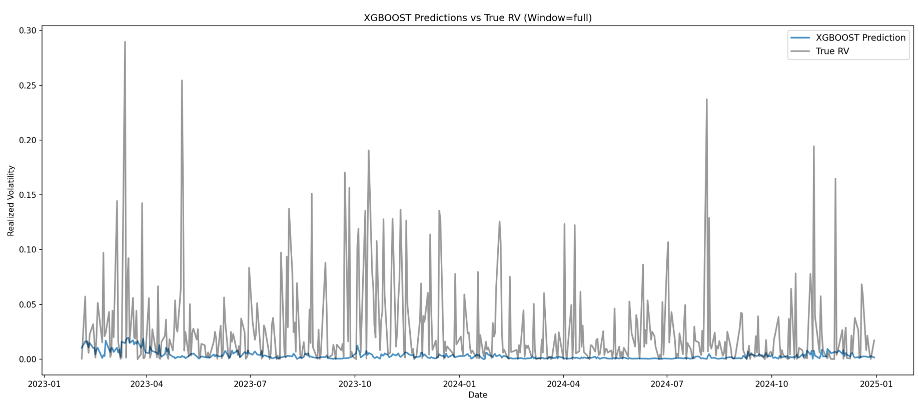

Figure 68: XGBOOST: Predictions vs True RV (Window=full)

XGBOOST Loss Metrics Over Time QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude for the ML model across different training periods.

XGBOOST QLIKE Loss Over Time (Window=full)

XGBOOST MSPE Loss Over Time (Window=full)

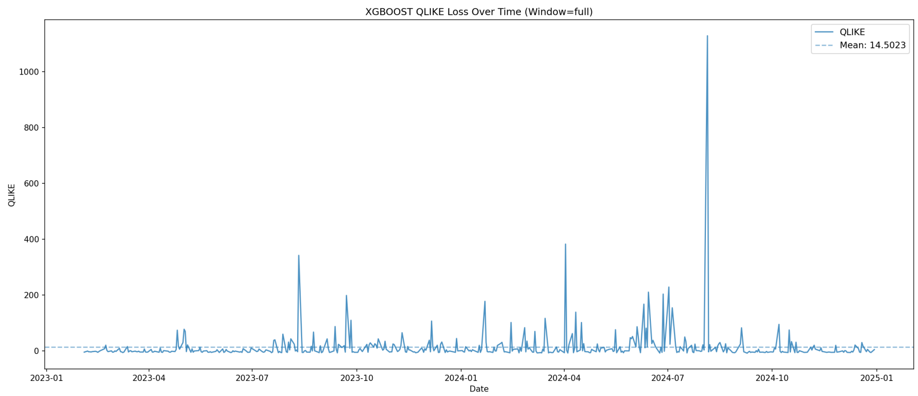

Figure 69: XGBOOST QLIKE Loss Over Time (Window=full)

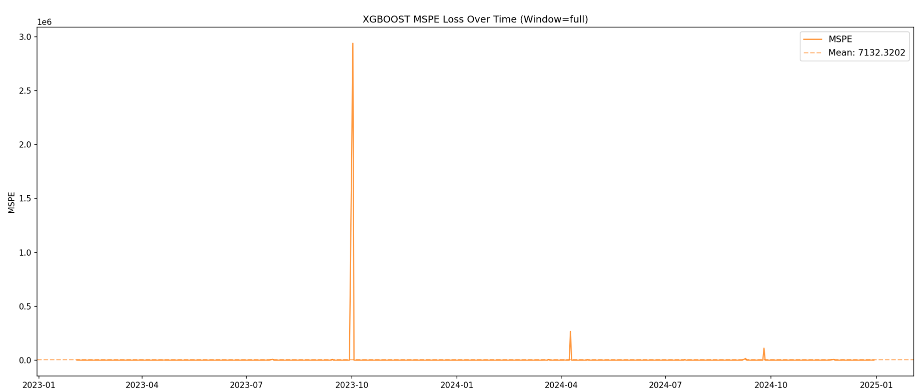

Figure 70: XGBOOST MSPE Loss Over Time (Window=full)

LIGHTGBM Model Performance

LightGBM (Light Gradient Boosting Machine) is a fast, distributed gradient boosting framework with GPU support. It uses leaf-wise tree growth for better accuracy and is especially efficient with large feature sets.

Table: LIGHTGBM Performance Summary Across Windows (Part 1/2)

	Window	QLIKE	QLIKE Std	MSPE
0	full	9.429	31.96	4642.25

Table: LIGHTGBM Performance Summary Across Windows (Part 2/2)

	MSPE Std	RMSE	Dir. Accuracy
0	66008.1	0.045	49.68%

LIGHTGBM Predictions vs True RV The following plots compare predicted volatility against true realized volatility for each rolling window.

LIGHTGBM: Predictions vs True RV (Window=full)

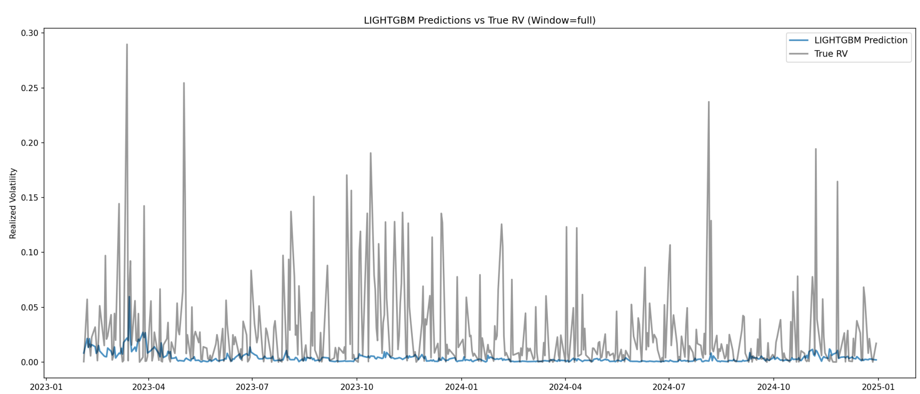

Figure 71: LIGHTGBM: Predictions vs True RV (Window=full)

LIGHTGBM Loss Metrics Over Time QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude for the ML model across different training periods.

LIGHTGBM QLIKE Loss Over Time (Window=full)

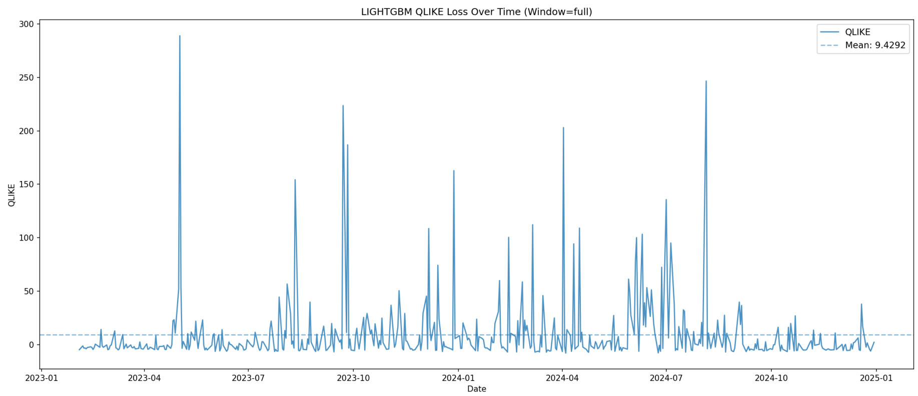

Figure 72: LIGHTGBM QLIKE Loss Over Time (Window=full)

LIGHTGBM MSPE Loss Over Time (Window=full)

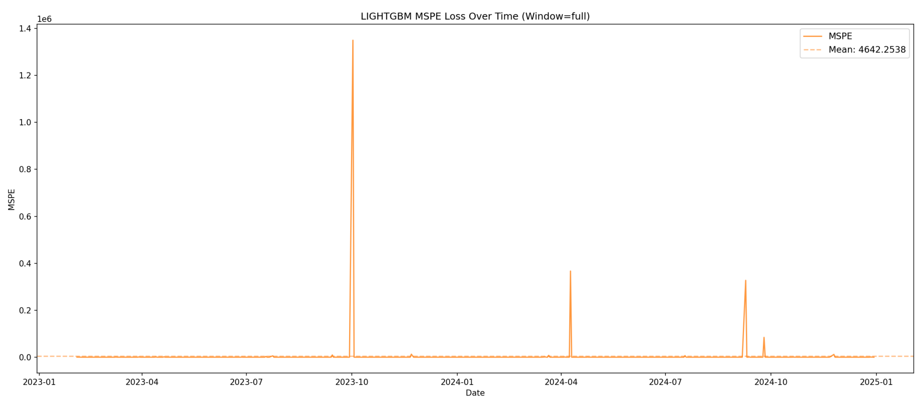

Figure 73: LIGHTGBM MSPE Loss Over Time (Window=full)

CATBOOST Model Performance

CatBoost (Categorical Boosting) is designed to handle categorical features effectively with GPU acceleration. It uses ordered boosting and symmetric trees to reduce overfitting, making it ideal for heterogeneous financial data.

Table: CATBOOST Performance Summary Across Windows (Part 1/2)

	Window	QLIKE	QLIKE Std	MSPE
0	full	2.108	15.197	18453.7

Table: CATBOOST Performance Summary Across Windows (Part 2/2)

	MSPE Std	RMSE	Dir. Accuracy
0	312392	0.045	50.53%

CATBOOST Predictions vs True RV The following plots compare predicted volatility against true realized volatility for each rolling window.

CATBOOST: Predictions vs True RV (Window=full)

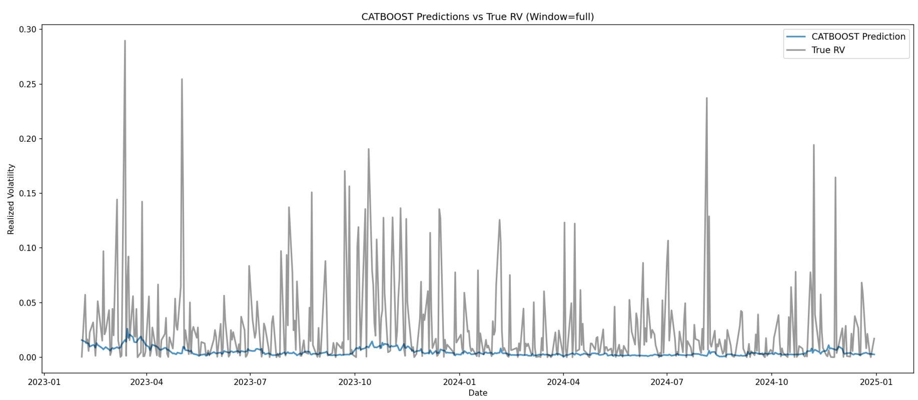

Figure 74: CATBOOST: Predictions vs True RV (Window=full)

CATBOOST Loss Metrics Over Time QLIKE (Quasi-Likelihood) and MSPE (Mean Squared Prediction Error) are computed over time for each window. These metrics help assess forecast calibration and error magnitude for the ML model across different training periods.

CATBOOST QLIKE Loss Over Time (Window=full)

CATBOOST MSPE Loss Over Time (Window=full)

ML Models Analysis Completed Successfully

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Temporal Fusion Transformer (TFT) Results

The Temporal Fusion Transformer is a state-of-the-art deep learning architecture for multi-horizon time series forecasting. It combines:

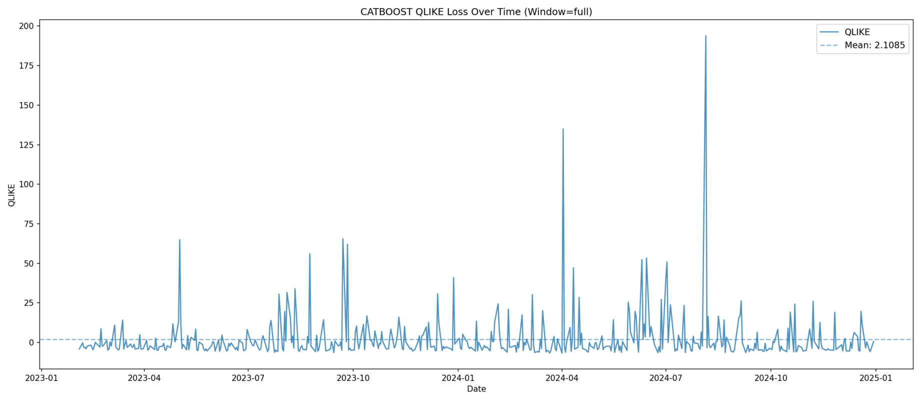

Figure 75: CATBOOST QLIKE Loss Over Time (Window=full)

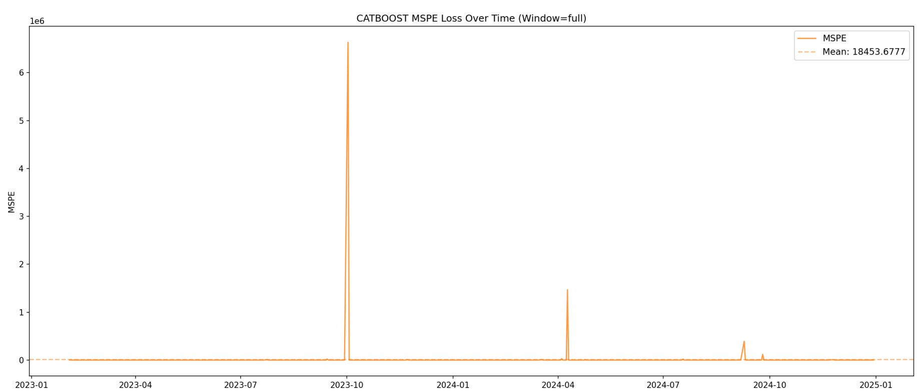

Figure 76: CATBOOST MSPE Loss Over Time (Window=full)

- **Multi-head attention mechanism**: Captures complex temporal dependencies
- **Variable selection networks**: Automatic feature importance learning
- **Gated residual networks**: Non-linear processing with skip connections
- **Quantile forecasting**: Provides prediction intervals (10th, 50th, 90th percentiles)

Key Improvements Implemented: - **Multiple quantiles**: Generates prediction intervals for uncertainty quantification - **Increased model capacity**: Larger hidden sizes and attention heads for better learning - **Enhanced regularization**: Higher dropout to prevent overfitting - **Extended lookback**: 90-day encoder length for capturing longer-term patterns

TFT Architecture Details: - Hidden size: 64 (optimized for dataset size) - Attention heads: 4 - Encoder length: 90 days (quarterly lookback) - Dropout: 0.1 (enhanced regularization) - Quantiles: 0.05, 0.1, 0.25, 0.5, 0.75, 0.9, 0.95 (7 quantiles) - Early stopping: Patience of 20 epochs

TFT Model Performance (Validation Set)

Metric	Value
QLIKE Mean	-2.533680
QLIKE Std	1.219096
QLIKE Min	-5.174289
QLIKE Max	7.203637
MSPE Mean	1569516.258582
MSPE Std	24253059.812042
MSPE Min	0.000001
MSPE Max	526681836.794077
RMSE	0.047893
Directional Accuracy	0.4406 (44.06%)
Training Samples	4523
Validation Samples	530

Figure: TFT Main Performance Analysis (Validation Set)

Figure: TFT QLIKE Distribution Analysis

Figure: TFT MSPE Distribution Analysis

Figure: TFT RMSE Analysis

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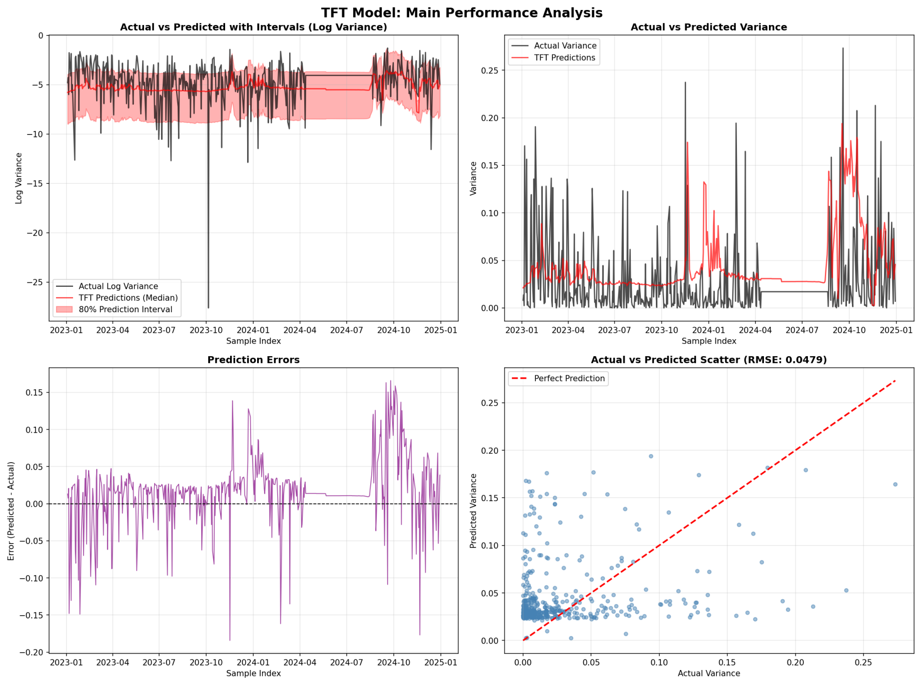

Figure 77: Figure: TFT Main Performance Analysis (Validation Set)

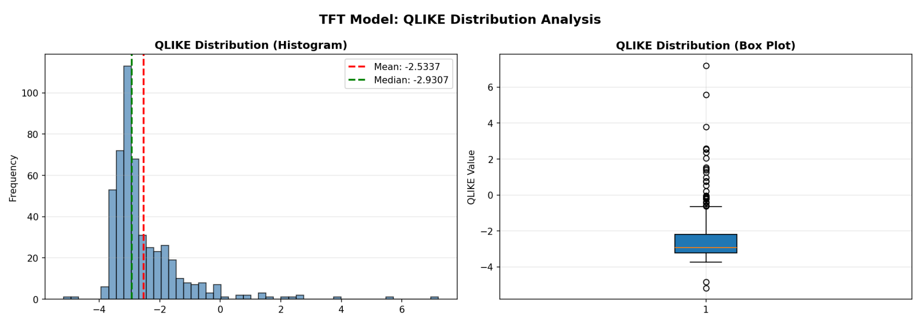

Figure 78: Figure: TFT QLIKE Distribution Analysis

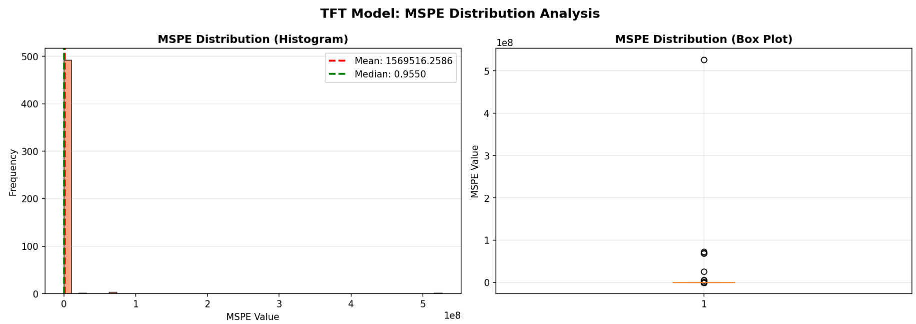

Figure 79: Figure: TFT MSPE Distribution Analysis

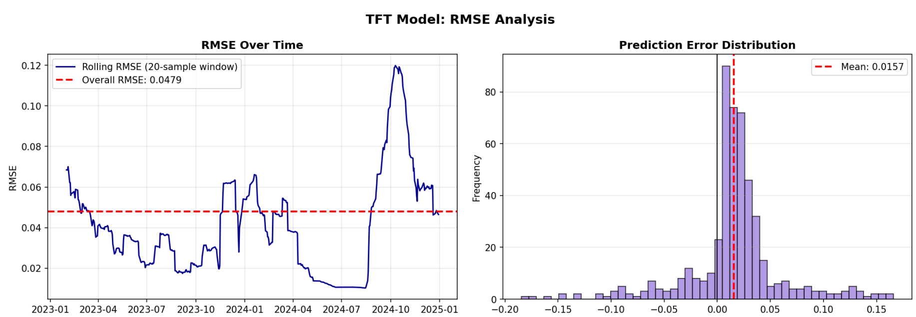

Figure 80: Figure: TFT RMSE Analysis